Forecast-aware latent modeling for longitudinal prediction of pathological complete response from breast MRI

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Abstract

Longitudinal breast MRI can reveal treatment response, but endpoint classifiers do not explicitly model the future trajectory implied by an observed history. We investigate forecast-aware latent modeling for pathological complete response (pCR) prediction using three-phase dynamic contrast-enhanced MRI, baseline clinical variables, and assigned-arm scenarios. An autonomous predictor estimates future latent states, and an attention readout combines predicted terminal state, change, and direct history features. The cohort comprises 976 patients and 3,641 examinations, with 801 development patients and a fixed internal holdout of 175 patients, including 56 positive outcomes. Three principal variants distinguish prediction without future-state supervision, auxiliary-supervised direct prediction, and forecast-attention readout. Every variant uses training seed 44 and five development folds. We evaluate each fold model separately and average its metrics; probabilities are not combined between models. Forecast attention achieved fold-mean T0/T1/T2 AUROCs of 0.7294/0.7381/0.7421 and Macro AUROC/AP/F1 of 0.7365/0.5990/0.5469, compared with 0.7219/0.5792/0.5437 for the control without future-state supervision. Its Macro AUROC fold standard deviation was 0.0159. Auxiliary-supervised direct readout had higher Macro F1 (0.5539). None of the nine principal paired comparisons met the Holm-adjusted significance criterion. Forecast attention's higher sensitivity accompanied lower specificity, worse probability-quality point estimates, and threshold-dependent hypothetical utility. These exploratory results require independent validation because seed selection and metric aggregation were retrospective, and repeated folds do not provide independent clinical replications.

Keywords: Breast MRI, Pathological complete response, Longitudinal imaging, Latent dynamics, Treatment response prediction

1. Introduction

Neoadjuvant systemic therapy provides an opportunity to monitor breast cancer before surgery and to study how an individual tumor responds over time. Pathological complete response (pCR) is a clinically meaningful endpoint associated with long-term outcomes, although its relationship to prognosis varies across biological subtypes and it should not be equated with a universally valid trial-level surrogate (Cortazar et al., 2014). Predicting pCR from imaging could support response assessment and the design of future adaptive studies. Such a prediction remains a prognostic estimate under an observed treatment context; it does not establish that changing treatment would improve an individual patient's outcome.

Dynamic contrast-enhanced MRI captures three-dimensional anatomy and contrast uptake patterns that evolve during therapy. Prior work in the I-SPY setting has demonstrated associations between MRI-derived tumor measurements and pathological response or subsequent outcomes (Hylton et al., 2012, 2016). Public multicenter imaging resources also make it possible to examine learned representations across repeated visits (Li et al., 2022; Newitt et al., 2021). However, a longitudinal prediction system must operate at the time a decision is considered: a baseline prediction can use baseline information, whereas an interim prediction can use the examination history accumulated by that interim visit. Future scans may inform representation learning on development patients but cannot become inputs to predictions for holdout patients at an earlier landmark.

Most direct classifiers optimize the endpoint of interest from observed images and clinical variables. This is a reasonable objective, but it does not explicitly require the representation to capture how the tumor state may evolve. A different strategy is to predict future imaging information and use the prediction to inform the endpoint readout. Image synthesis provides one route to such a representation, but prediction in a compact latent space can focus the training signal on state evolution without requiring a pixel-space image generator at inference. This distinction is useful for response prediction: the relevant question is whether the learned forecast contributes information to classification, rather than whether a generated image is visually convincing.

Joint-embedding predictive learning offers a general

mechanism for predicting target representations computed by a slowly updated encoder (Assran et al., 2023). In a longitudinal medical setting, adapting that principle requires explicit handling of partial histories, missing future observations, treatment context, and the distinction between an auxiliary training target and a classifier input. A model may benefit from a future-state loss while still making a direct prediction from the observed history. Conversely, directly including forecast features in the classifier changes both its information pathway and its parameterization. These effects should be examined through matched controls rather than attributed to a single generic "world model" component.

We investigate forecast-aware latent modeling for longitudinal pCR prediction from breast MRI. The model encodes three selected DCE phases at each available visit, fuses observed MRI with baseline clinical information and an assigned treatment-arm scenario, and autonomously rolls the current latent state forward to a terminal stage. A spatial attention mechanism summarizes the predicted terminal state and the predicted change from the current state for pCR classification. We compare the complete model with two direct-readout controls: one without future-state supervision and one with the same auxiliary supervision as the complete model. A supplementary mean-pooling model further examines the attention component.

The study makes three concrete contributions. First, it formulates a common representation and masked future-state objective that can be applied to available T0–T2 histories while using observed later scans only as development-time targets. Second, it provides a staged comparison that distinguishes the effect of auxiliary prediction from the joint effect of forecast readout and attention. Third, it reports landmark-specific discrimination, class-specific errors, calibration, hypothetical decision curves, subgroup heterogeneity, archived forecasting diagnostics, and paired patient-level uncertainty on a fixed internal holdout. The manuscript evaluates each variant with training seed 44. Performance metrics are computed separately for each of five fold-specific models and summarized by their mean and descriptive sample standard deviation. Patient-level inference preserves the dependence between repeated predictions on the same internal holdout. The seed and aggregation policy were chosen retrospectively, and upstream-independence limits remain explicit.

2. Related work

MRI biomarkers and learned pCR prediction.. MRI-based response assessment has a substantial clinical and quantitative foundation. Functional tumor volume and related measurements have been studied in longitudinal breast MRI cohorts, including I-SPY (Hylton et al., 2012, 2016). The ACRIN 6698 study additionally examined diffusion-derived measurements in relation to response (Partridge et al., 2018). These studies motivate imaging-based monitoring but do not establish that a learned model using different inputs or a different evaluation split has equivalent clinical validity. Our experiments use selected DCE phases and baseline clinical variables; diffusion imaging is not an input to the present model.

Learning-based approaches can combine complementary image sequences or repeated examinations. Synthesis-based representation learning has been explored for multi-sequence 3D/4D MRI (Han et al., 2024), and recent studies have examined spatiotemporal or early-treatment information for pCR prediction (Tang et al., 2025; Du et al., 2025). Such studies provide relevant methodological context, but differences in patient selection, imaging availability, endpoints, and holdout construction prevent direct numerical comparisons with our results. We therefore do not claim superiority to published systems from cross-paper metric comparisons. Reimplementation of suitable external baselines under the same patient split remains necessary.

Predictive latent representations.. Predicting a target embedding rather than reconstructing the input image is a central idea in joint-embedding predictive learning (Assran et al., 2023). We use an exponential-moving-average target encoder to define latent states and supervise autonomous future predictions. The present objective is a supervised pCR loss combined with a masked multi-step latent-state loss. It does not activate a separate single-step teacher-forcing JEPA loss, image reconstruction loss, or latent-variance regularizer. This objective-level distinction matters when interpreting the internal experiment name NoJEPA: that control removes future-state supervision, while retaining the common fusion architecture.

Attention and volumetric image representation.. Transformer attention provides a flexible way to combine image tokens and contextual information (Vaswani et al., 2017). Compact convolutional representations can reduce the cost of processing volumetric MRI, and vector-quantized generative models provide one source of pretrained convolutional encoders (Esser et al., 2021). Here the pretrained codec contributes a continuous convolutional encoder. The quantizer and image decoder are bypassed, and the main experiments do not synthesize MRI volumes. The final attention weights summarize latent tokens that have already undergone spatial mixing; they are therefore not validated lesion segmentations or anatomical explanations.

3. Method

3.1. Longitudinal prediction problem

We consider prediction of pathological complete response (pCR) from a patient's available magnetic resonance imaging (MRI) history during neoadjuvant treatment. Canonical examination stages are indexed by $t \in \{0, 1, 2, 3\}$. At a prediction landmark $k \in \{0, 1, 2\}$, the model receives only examinations acquired and available by that landmark. Let $X_t \in \mathbb{R}^{3\times32\times128\times128}$ denote the processed examination at stage t, and let $o_t^{(k)}$ indicate whether this examination

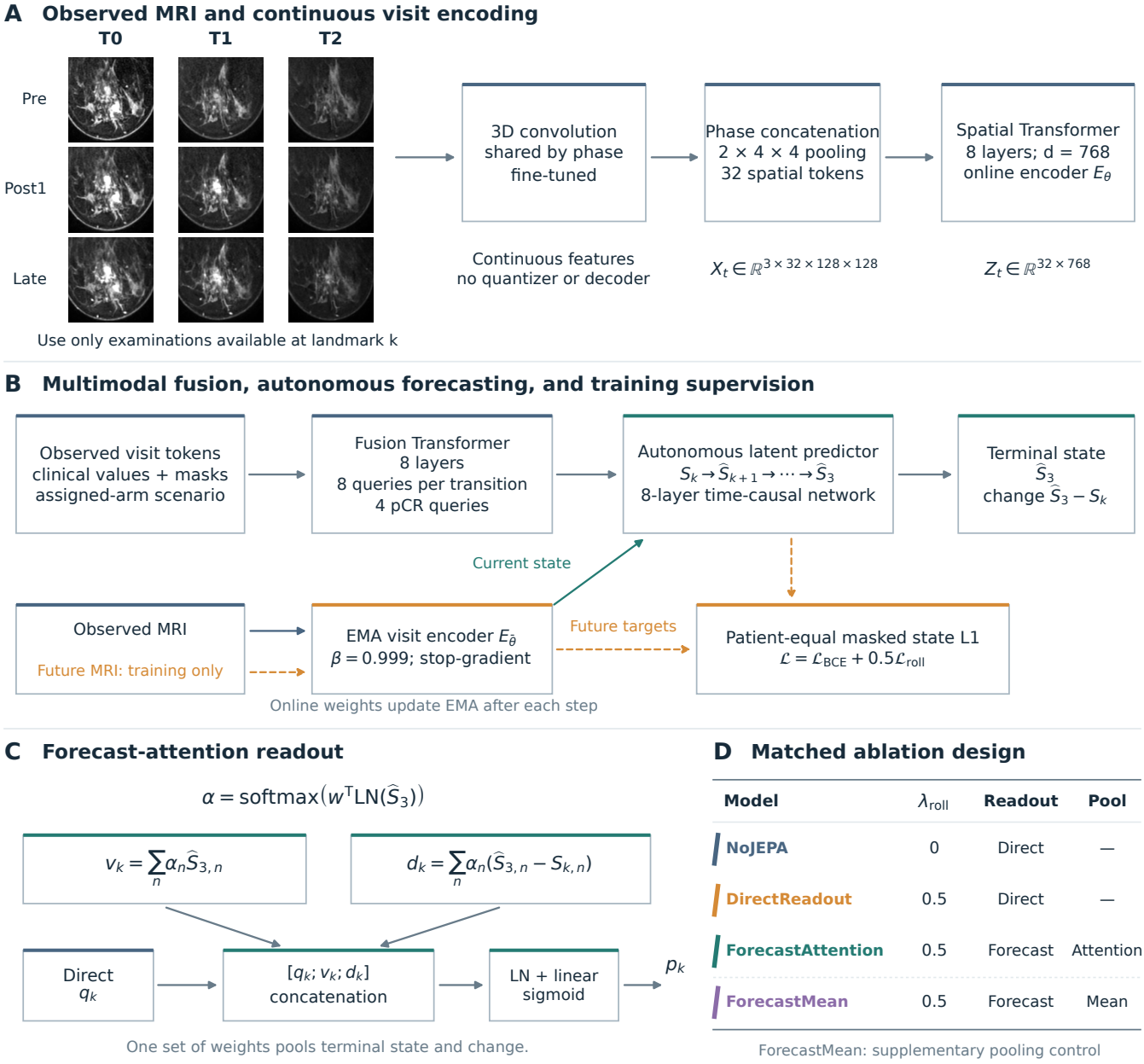


Figure 1: Forecast-aware latent modeling and matched controls. (A) Real three-phase MRI crops from one development patient illustrate the same baseline-defined plane at T0, T1, and T2; the example and slice were selected without outcomes or pCR predictions. Only visits available at the prediction landmark are inputs. A shared continuous encoder and spatial Transformer produce visit tokens. (B) Clinical values, availability masks, and an assigned-arm scenario condition fusion and autonomous prediction to T3. The EMA encoder supplies the detached current state; available future MRI supplies training targets only. Dashed orange arrows denote training-target or loss paths. (C) Attention weights pool the predicted terminal state and its change; these features are combined with direct pCR-query features. (D) The primary variants separate removal of future-state supervision from the complete forecast readout. ForecastMean is a supplementary pooling control. One seed44 fold model is illustrated. Each of the five fold models is evaluated separately on the same patients, and reported performance averages the five separately computed fold metrics. Prediction probabilities are never averaged. MRI tiles are illustrative crops, not generated images, segmentation ground truth, or attention maps.

is available in the observed prefix. Missing examinations retain their canonical stage rather than shifting later examinations forward. The three channels contain the pre-contrast acquisition, the first post-contrast acquisition, and a metadata-selected late post-contrast acquisition, respectively. The late acquisition is not assumed to have the same acquisition index for all examinations.

The observed information is

$$\mathcal{H}_k = \{(X_t, t) : o_t^{(k)} = 1\} \cup \{c_k, m_k, a_k, \tau_k\}, \qquad (1)$$

where $c_k \in \mathbb{R}^4$ contains the available clinical variables, m_k their availability indicators, a_k the available assigned

treatment-arm category, and τ_k its declared availability stage. The clinical variables are age at screening, hormone-receptor positivity, HER2 positivity, and a binary MammaPrint variable. Age is standardized using the current training partition; binary variables retain their original coding. Unavailable values are set to zero after standardization and accompanied by an explicit mask. The clinical token embeds the concatenated values and masks. Treatment conditioning uses 13 released arm categories and an additional unknown category. In this retrospective experiment, assigned arm and available baseline clinical variables are assumed to be known at T0. This is an assigned-arm scenario; the categorical input does not specify actual doses, treatment intervals, or treatment changes, and it does not support a causal interpretation.

The output is $p_k = \Pr(y = 1 \mid \mathcal{H}_k)$, where $y \in \{0, 1\}$ is the final pCR label. Future MRI is stored separately as training supervision and is excluded from the prediction input. T3 can provide a future representation target during training, but is never a prediction landmark in the reported T0–T2 protocol.

3.2. Continuous MRI encoding and multimodal fusion

Each observed examination is encoded independently by an online visit encoder E_θ. Its convolutional front end shares the same single-channel weights across the three acquisition phases. Two stride-two three-dimensional convolutional stages, residual blocks, and group normalization produce 64 continuous feature channels per phase at one-quarter input resolution along each spatial axis. The three phase-specific feature maps are concatenated, adaptively averaged to a $2 \times 4 \times 4$ grid, and projected to a latent width of $d = 768$. This gives $N = 32$ spatial tokens per examination. Learned spatial positional embeddings and an eight-layer spatial Transformer then produce the visit representation $Z_t = E_\theta(X_t) \in \mathbb{R}^{N \times d}$.

The convolutional front end is initialized from the same pre-existing VQ-GAN-style codec checkpoint for all compared models and is fine-tuned. Here, the model uses continuous features from the codec encoder before its compression projection. It does not use discrete codebook assignments, a quantization objective, a decoder, or adversarial learning during downstream pCR training. [TO COMPLETE: Document the original codec pretraining dataset, patient-level overlap with the present partitions, checkpoint provenance, and applicable redistribution terms.]

Stage embeddings distinguish examination times. Observed image tokens, one clinical token, and one treatment-arm token form the multimodal context. An eight-layer fusion Transformer additionally receives eight learned transition-query tokens for each of the three canonical transitions and four pCR-query tokens. Transition-query slots are retained at every landmark, independently of which future targets happen to be available. Their conditioning is derived entirely from the observed prefix. Context tokens interact bidirectionally; appended transition and pCR queries use a causal attention mask and can attend to the context and preceding query positions. Padding masks remove unobserved image tokens. This produces transition features $A_{j+1}^{(k)} \in \mathbb{R}^{8 \times d}$ for transitions $j \to j + 1$, and pCR-query features $Q_k \in \mathbb{R}^{4 \times d}$.

All spatial, fusion, and prediction Transformers use pre-normalization blocks, GELU feed-forward layers of width 3072, and dropout 0.3, following the attention-based architecture of Vaswani et al. (2017). Attention has 16 heads with 64 channels per head: its internal width is therefore 1024, followed by projection back to the 768-dimensional residual stream. The clinical and treatment tokens condition forecasting through the shared fusion network rather than through a separately supplied future treatment trajectory.

3.3. EMA target states and autonomous forecasting

A target visit encoder $E_{\bar\theta}$ is initialized as a copy of the online encoder and updated after each optimizer step by exponential moving average (EMA):

$$\bar\theta \leftarrow \beta\bar\theta + (1 - \beta)\theta, \qquad \beta = 0.999. \tag{2}$$

The target encoder remains in evaluation mode and receives no gradients. It defines a common representation space for the observed initial state and available future targets,

$$S_t = \mathrm{stopgrad}(E_{\bar\theta}(X_t)) \in \mathbb{R}^{N \times d}. \tag{3}$$

Encoding a future target in this branch supplies a loss target only; it does not make that examination available to the forecasting or classification input.

The predictor starts from the latest actually observed state in $\mathcal{H}_k$ and recursively advances to T3. For the reported landmarks, this initial state is S_k. At each future transition, it uses the corresponding transition features and its previously generated states. A history input additionally embeds the difference between a source state and its latest available predecessor, with a separate indicator for whether a predecessor exists. Observed history is used where available; after the landmark, the history is extended using model predictions. Thus, a missing future examination removes supervision at that time but does not interrupt the simulated transition sequence.

The prediction network is an eight-layer time-causal Transformer. Each transition block contains its eight transition tokens and 32 spatial source-state tokens, together with learned spatial and temporal embeddings. Tokens can interact within a block; attention between transition blocks is directed from earlier to later blocks. A learned root token remains available when some blocks are masked. The network predicts a residual state increment,

$$\widehat{S}_{j+1} = \widetilde{S}_j + G_{\phi,j}\left(\widetilde{S}_{k:j}, A_{k+1:j+1}^{(k)}, D_{k:j}\right), \qquad j = k, \ldots, 2, \tag{4}$$

where $\widetilde{S}_k = S_k$, subsequent $\widetilde{S}_j$ are generated states, and $D_{k:j}$ denotes the available predecessor differences and validity indicators. The residual output projection is initialized

to zero, so prediction initially preserves the source state. The transition features summarize the complete legal prefix even when only a subset of previous state differences is available to a particular prediction block.

Forecasts are generated autonomously during training and inference. A true future state never replaces a predicted predecessor. The classifier likewise uses this autonomous trajectory. The configuration includes no teacher-forced next-state loss in these experiments. EMA encoding of real future examinations is used exclusively to evaluate representation error against available targets.

3.4. Prediction-conditioned pCR readout

The full model, ForecastAttention, combines a direct summary of the observed prefix with summaries of the predicted terminal state and its change from the current state. First, the four pCR-query features are averaged to obtain $q_k = \frac{1}{4}\sum_{r=1}^{4} Q_{k,r}$. For spatial token n, the forecast change is $\Delta S_{k,n} = \widehat{S}_{3,n} - S_{k,n}$. A learned scalar scorer defines

$$\alpha_{k,n} = \frac{\exp\left(w^\top \mathrm{LN}(\widehat{S}_{3,n})\right)}{\sum_{u=1}^{N} \exp\left(w^\top \mathrm{LN}(\widehat{S}_{3,u})\right)}. \tag{5}$$

The same weights pool terminal state and change,

$$v_k = \sum_{n=1}^{N} \alpha_{k,n} \widehat{S}_{3,n}, \qquad d_k = \sum_{n=1}^{N} \alpha_{k,n} \Delta S_{k,n}. \tag{6}$$

The final prediction is

$$p_k = \sigma\left(b + u^\top \mathrm{LN}([q_k; v_k; d_k])\right). \tag{7}$$

The scorer is initialized with $w = 0$, corresponding to uniform pooling at initialization. Classification gradients propagate through the generated terminal state, predictor, and transition-conditioning pathway. The observed EMA reference state remains detached. The attention weights summarize transformed spatial tokens and should not be interpreted as validated anatomical localization or tumor segmentation.

For DirectReadout, pCR is instead predicted directly from q_k using a layer-normalized linear head. Its future-state predictor is trained as an auxiliary task, but generated states are not explicit inputs to the pCR head. The direct head has input width d, whereas the full readout has width $3d$. Their comparison therefore evaluates the complete forecast-conditioned readout, including its state and change inputs and classifier width. An additional mean-pooling variant, when reported, retains the forecast-conditioned readout and replaces $\alpha_{k,n}$ by $1/N$.

3.5. Training objective and matched ablations

Let $\mathcal{B}_y$ be the labeled patients in a training batch, and let $m_{i,t}^{\text{future}}$ indicate an actually available future target for patient i after its sampled landmark. The pCR objective is unweighted binary cross-entropy,

$$\mathcal{L}_{\text{pCR}} = -\frac{1}{|\mathcal{B}_y|} \sum_{i \in \mathcal{B}_y} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right]. \tag{8}$$

Define $M_i = \sum_t m_{i,t}^{\text{future}}$ and $\mathcal{B}_+ = \{i : M_i > 0\}$. Autonomous representation supervision is patient-equal:

$$\mathcal{L}_{\text{roll}} = \frac{1}{|\mathcal{B}_+|} \sum_{i \in \mathcal{B}_+} \frac{1}{M_i} \sum_t m_{i,t}^{\text{future}} \frac{\|\widehat{S}_{i,t} - S_{i,t}\|_1}{Nd}. \tag{9}$$

If a batch contains no future targets, this term is zero. Each available future examination has equal weight within a patient; the loss uses no horizon discount, lesion weighting, or fabricated target for a missing visit. Patients without future MRI still contribute pCR supervision.

The full model and DirectReadout minimize

$$\mathcal{L} = \mathcal{L}_{\text{pCR}} + 0.5\mathcal{L}_{\text{roll}}. \tag{10}$$

NoJEPA uses the direct pCR head with $\mathcal{L} = \mathcal{L}_{\text{pCR}}$. This historical control name denotes removal of future-state auxiliary supervision, rather than removal of every transition-related module: it retains the shared fusion architecture and transition-query slots. Consequently, DirectReadout versus NoJEPA isolates the added autonomous representation objective under the same direct readout, whereas ForecastAttention versus DirectReadout assesses the added forecast-conditioned classification pathway. The latter comparison cannot isolate spatial attention alone.

All three models omit teacher-forced JEPA, image reconstruction, flow-matching, and variance-regularization losses. In particular, the future objective reported here is autonomous state prediction, despite the JEPA-related implementation lineage. During training, patients are shuffled once per epoch and one actual available T0–T2 landmark is sampled uniformly for each patient. This balances patient contributions without treating a patient with more visits as several independently sampled patients. The same clinical endpoint supervises each sampled prefix; validation assesses all available eligible prefixes using patient-equal weighting.

4. Dataset and preprocessing

4.1. Cohort construction and clinical variables

We retrospectively analysed longitudinal breast dynamic contrast-enhanced MRI (DCE-MRI) from the public I-SPY2 and ACRIN-6698 imaging resources Li et al. (2022); Newitt et al. (2021). Source imaging identifiers were linked to research identifiers using the accompanying official identity mapping, and outcomes and baseline clinical variables were obtained from the official I-SPY2 Imaging Cohort 1 clinical spreadsheet. The final cohort comprised 976 patients with 3,641 usable examinations. Native imaging

identifiers belonged to the ISPY2 and ACRIN-6698 namespaces for 712 and 264 patients, respectively; these identifiers were mapped to the same clinical release and do not define independent development and external validation cohorts. The official collection pages identify the source imaging and clinical release as CC BY 4.0. [TO COMPLETE: Confirm terms for redistributing derived assets and document the applicable ethics approval or waiver and original consent arrangements.]

Eligibility required an actual baseline examination, an available binary pCR label, usable baseline registered image assets, and a nonempty baseline localisation produced by the existing MAMA-MIA segmentation workflow. Of 980 clinically matched source candidates, three lacked a baseline examination and one had an empty or unavailable predicted baseline localisation. Their exclusion yielded 976 patients and 3,642 candidate examinations. One post-treatment examination was subsequently marked unavailable because its saved within-examination phase transforms displaced the postcontrast images outside the prescribed baseline crop. The patient's earlier examinations were retained, leaving 3,641 usable examinations. Missing follow-up examinations did not cause patient exclusion. No additional admission thresholds based on normalised cross-correlation, field-of-view ratios, functional tumour volume, or the availability of an accepted adjacent longitudinal pair were applied.

The endpoint was the released binary pCR field: 313 patients were labelled positive and 663 negative. [TO COMPLETE: Verify and state the release-specific pathological definition of pCR, including breast and nodal criteria and the handling of in-situ disease; do not infer this definition from the binary column alone.] Baseline covariates were age at screening, hormone-receptor (HR) status, HER2 status, and the released binary MammaPrint indicator. Age was available for 973 patients; all three binary covariates were available for the entire cohort. The official assigned treatment arm, represented by 13 categories, was also supplied to the model. The current experiments use an *assigned-arm scenario*: the assigned arm and available baseline covariates are assumed known at baseline. The spreadsheet does not establish prospective allocation timestamps, doses, cycle-level exposure, or interval treatment schedules, and the model does not reconstruct these quantities.

Examinations were indexed by the canonical study stages T0–T3 rather than continuous elapsed time. Availability at these stages was 976, 935, 875, and 855 patients, respectively (Table 1); 803 patients had all four examinations and 861 had T0–T2. The nominal source protocol acquired T0 before randomization, T1 after three weeks of the first regimen (paclitaxel with or without an experimental agent), T2 between paclitaxel and anthracycline regimens, and T3 after four cycles of anthracycline treatment and before surgery (Li et al., 2022). Because T0 MRI preceded randomization, supplying the eventual assigned arm at T0 is a retrospective conditional scenario, rather than a prospectively available predictor before treatment allocation. The source protocol used 1.5 or 3.0 Tesla scanners with dedicated breast coils and required the same scanner configuration within a patient. These are nominal protocol specifications, not a patient-level audit of timing or scanner compliance. Twenty-three patients had T0 only and remained eligible for pCR supervision. Their unavailable future MRI targets were masked, and no synthetic follow-up images were substituted. Model inputs and evaluation landmarks were restricted to T0, T1, and T2; a real T3 examination, when available in the training partition, could provide a future-state target. [TO COMPLETE: Report the actual examination intervals, acquisition centres, scanner distribution, and sequence-parameter ranges for the final analysed cohort.]

4.2. Patient partitioning and evaluation scope

The cohort was divided into 801 development patients and a fixed internal holdout of 175 patients, including 56 pCR-positive patients. All holdout patients had actual T0, T1, and T2 examinations, allowing matched longitudinal evaluation; 163 also had T3. The holdout retained 139 patients with complete T0–T2 observations from the historical benchmark and added 36 eligible patients. Additional patients were selected using outcome quotas and a fixed baseline-clinical distribution balancing rule with split seed 2026, without using model predictions or performance. This fixed cohort-construction seed is distinct from the downstream training seed 44; the patient split is unchanged. Development folds were stratified by pCR and examination-availability pattern; rare joint strata were combined within the same pCR class. Each development patient appeared in validation exactly once across the five folds. Fold 1 contained 640 training and 161 validation patients; the remaining folds each contained 641 training and 160 validation patients. Training, validation, and holdout identities were disjoint within each fold, and the same holdout was used for every fold model trained with seed 44.

This partition is an internal downstream benchmark. In particular, 29 of the 36 added holdout patients had belonged to the development population of an earlier experiment, so earlier trained models cannot be treated as held out on the entire new set. The present downstream models were trained using the new fold definitions. Patient independence of the supplied codec and segmentation pretraining checkpoints relative to the final holdout has not yet been established; that provenance must be audited before stronger claims of end-to-end independence. The complete-examination holdout eligibility also restricts inference about patients with missing early follow-up examinations.

4.3. Registration, fixed localisation, and normalisation

Within each examination, postcontrast phases were registered to the precontrast reference. Follow-up examinations were aligned to the patient's T0 reference using the saved longitudinal transforms, retaining a rigid fallback when the final deformable candidate was not usable. Baseline localisation used the existing MAMA-MIA nnU-Net prediction

Table 1: Characteristics of the analysed cohort. Development statistics count each of the 801 development patients once, rather than pooling overlapping training folds. Age is reported as mean ± sample standard deviation among patients with observed age; other entries are counts.

Characteristic	All	Development	Holdout
Patients	976	801	175
pCR positive	313	257	56
pCR negative	663	544	119
Age (years)	48.7 ± 10.5	48.7 ± 10.6	48.9 ± 10.2
Missing age	3	3	0
HR positive	533	434	99
HER2 positive	240	195	45
Available T0	976	801	175
Available T1	935	760	175
Available T2	875	700	175
Available T3	855	692	163

on the native first-postcontrast examination Garrucho et al. (2025). The centre of the largest 26-connected predicted component defined a fixed baseline crop, which was reused at every actual follow-up. Follow-up segmentation and follow-up-driven recentering were not used.

Each examination was represented by three channels: the precontrast phase, the first postcontrast phase, and the late phase explicitly specified in the examination metadata. The late channel was not selected by assuming a fixed phase number or taking the last available phase. Images were resampled to $32 \times 128 \times 128$ voxels with spacing $2.0 \times 0.7032 \times 0.7032$ mm in the slice and two in-plane dimensions, approximately covering $64 \times 90 \times 90$ mm. Linear interpolation was used for MRI intensities and nearest-neighbour interpolation for masks and acquisition support, with zero padding outside the acquired field of view. The localisation mask determined the crop; it was not a contemporaneous ground-truth tumour segmentation or a future observation supplied to the classifier.

Native metadata, baseline mask headers, registered reference grids, and saved transforms were maintained as matched versions. Explicit array-index mappings, rather than placeholder NIfTI affines, defined sampling positions. The exported arrays were checked for finite values, consistent coordinate mapping, and numerical agreement after compressed storage and readback. These checks establish data and geometry consistency but do not establish anatomical registration or segmentation accuracy. The fixed crop partly truncated the largest predicted baseline component for 85 patients; these patients were retained and the limitation was recorded.

Raw cropped intensities were stored once and shared across folds. For each fold, a single intensity mean and population standard deviation were fitted to nonzero precontrast foreground voxels from every unique actual examination of that fold's training patients. The same scalar transform was applied once to the foreground of all three channels, stages, and partitions; background and padding remained zero. Validation and holdout images were excluded from fitting. Age normalisation was likewise fitted to observed training-patient values; HR, HER2, and MammaPrint retained their official binary coding, with missingness represented explicitly.

5. Experimental design

5.1. Development protocol and landmark evaluation

The 801 development patients were partitioned into five patient-level folds. Fold 1 used 640 training and 161 validation patients; folds 2–5 each used 641 training and 160 validation patients. The same 175-patient holdout was evaluated by every fold model and every variant. The five fold-specific test evaluations are repeated evaluations of models trained on overlapping development partitions and tested on the same patients. They do not constitute five independent clinical cohorts. Patient identities were disjoint between training, validation, and holdout within the current split.

At each training epoch, a patient contributed one randomly selected available T0, T1, or T2 prefix. Only MRI observations available at or before the selected landmark were classifier inputs. Later observed visits, including T3, could supply latent targets for the auxiliary loss on training patients. Missing targets were masked. The 23 patients with only T0 remained eligible for pCR training but supplied no future-state loss. At evaluation, all three landmarks were evaluated for the 175 holdout patients, each of whom had T0, T1, and T2. T3 was not an evaluation landmark and was not used as observed input for these predictions.

5.2. Model comparisons

The principal comparison comprises three variants trained with seed 44 using the same five development folds. Every fold model is evaluated separately. Performance is summarized by the equal mean of the five fold-specific metric values. NoJEPA uses direct pCR readout and a zero future-state loss weight. DirectReadout uses the same direct readout with a future-state loss weight of 0.5. ForecastAttention retains that auxiliary weight and replaces the direct readout with a concatenation of direct

pCR features, attention-pooled predicted terminal state, and attention-pooled predicted change. All variants retain the shared image encoder, clinical inputs, assigned-arm encoding, fusion architecture, and training schedule.

The DirectReadout–NoJEPA contrast most directly tests adding future-state supervision. The ForecastAttention–DirectReadout contrast jointly changes the use of forecast features, spatial pooling, and classifier input width. It cannot isolate attention alone. For that purpose, the supplementary ForecastMean comparison uses forecast readout with mean spatial pooling and otherwise matches the forecast-attention configuration. All manuscript-facing pCR results first compute metrics separately for each fold model on the fixed holdout and then average those metrics across folds. Probabilities and thresholded predictions are never averaged between fold models. No VQ-latent reconstruction-head, change-weighted-loss, or state-conditioned-head variants are included.

5.3. Implementation and optimization

The network uses a model width of 768, eight layers in each of the spatial encoder, multimodal fusion module, and predictor, 16 attention heads with 64 dimensions per head, and feed-forward width 3,072. The 1,024-dimensional concatenated attention-head representation is projected back to the 768-dimensional residual stream. The spatial token grid is $2 \times 4 \times 4$. The configuration uses eight state queries, four pCR queries, dropout 0.3, and an EMA coefficient of 0.999. The pretrained convolutional encoder is fine-tuned jointly with the remaining network. Its training-patient provenance requires additional documentation, as discussed in Section 7.5.

We use AdamW (Loshchilov and Hutter, 2019) with an initial base learning rate of 10^{-4} for both the encoder and the main network, weight decay 0.05, batch size 16, no gradient accumulation, and gradient-norm clipping at 1.0. Training uses BF16 mixed precision. A ten-epoch warm-up increases the learning rate from 10^{-5} to 10^{-4}, followed by cosine decay to 10^{-6} at epoch 100. Training is capped at 200 epochs, with early stopping after 50 epochs without improvement. The best checkpoint is selected independently in each fold using validation patient-equal-prefix F1 at threshold 0.5. This is an F1 selection protocol, even though AUROC is a principal reporting measure. Deterministic execution was not enabled, so a fixed seed does not imply bitwise reproducibility across hardware or software changes.

5.4. Metrics, aggregation, and patient-level uncertainty

We report 14 complementary metrics: AUROC, non-interpolated average precision (AP), F1, accuracy, sensitivity, specificity, positive and negative predictive value (PPV/NPV), balanced accuracy, Matthews correlation coefficient (MCC), Youden index, negative log-likelihood (NLL), Brier score, and expected calibration error (ECE) with ten equal-width probability bins. Confusion-matrix counts are additionally retained. Thresholded primary metrics use 0.5. MCC and balanced accuracy help interpret performance under class imbalance (Chicco and Jurman, 2020); positive prevalence is $56/175 = 0.32$. No holdout threshold optimization or probability recalibration was performed.

Let $p_{if\ell}$ be the probability from the seed-44 model trained in fold f for patient i at landmark ℓ. Within each variant, we define

$$\begin{aligned} M_{f\ell} &= M\big(\{y_i, p_{if\ell}\}_{i=1}^{175}\big), \\ M_{f,\text{macro}} &= \frac{1}{3}\sum_{\ell=0}^{2} M_{f\ell}, \\ \widehat{M}_\ell &= \frac{1}{5}\sum_{f=1}^{5} M_{f\ell}, \\ s_{M,\ell} &= \sqrt{\frac{1}{4}\sum_{f=1}^{5}(M_{f\ell} - \widehat{M}_\ell)^2}. \end{aligned} \tag{11}$$

Macro is first computed within each fold by averaging T0–T2 metric values, including F1, and is then averaged over folds. The corresponding Macro standard deviation is calculated from the five fold-specific Macro values. Main tables report the fold mean and sample standard deviation, with denominator four. This spread describes fitted-model differences associated with the five overlapping training partitions under seed 44. It is not a training-seed standard deviation, standard error, or patient-level interval. All probabilities remain specific to individual models; the reported mean is not the performance of a probability ensemble or of a model trained on the full development cohort.

For conditional patient-level uncertainty, we draw 2,000 bootstrap samples of the 175 patients with replacement. Every draw is shared across all fold models, variants, and landmarks. Within each draw, we recompute metrics separately for each fold model, average landmark metrics for Macro, and then equally average the five fold metrics. The 2.5th and 97.5th percentiles give unadjusted intervals for the fold-mean metric or paired difference. The fitted models are held fixed and folds are not resampled. These intervals exclude retraining, seed selection, model selection, and aggregation-policy uncertainty. Neither fold count nor repeated landmarks increase the independent patient count beyond 175.

ROC and precision–recall curves are computed separately for each fold model, interpolated onto common false-positive-rate and recall grids, respectively, and then averaged for display. The numerical AUROC and non-interpolated AP are computed per fold before averaging; their means need not equal the area under the displayed mean curve. Pointwise bands use shared patient resamples. Reliability diagrams form five equal-count rank groups of 35 patients separately within each fold model and then average group summaries across folds. These are descriptive summaries of several predictors, distinct from the ten

equal-width bins used to calculate each fold model's ECE.

5.5. Exploratory paired significance checks

We use approximate, two-sided, null-centered paired patient-bootstrap tests for all 14 metrics, including AUROC (Davison and Hinkley, 1997). The tested quantity is the difference in the metric averaged across the five fixed fold models. We generate $B = 99{,}999$ shared patient resamples with random-number seed 20261103. Every resample draws 175 patients with replacement and is shared across folds, models, and landmarks. Metrics are recomputed separately for each fold before equally averaging, preserving the repeated-patient dependence. No model is retrained, no fold is resampled, and no fold-level t test is used.

Let Δ_b^* denote the fold-mean metric difference in resample b and Δ_{obs} its observed value. We estimate the two-sided p value by

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{I}\big(|\Delta_b^* - \Delta_{\text{obs}}| \geq |\Delta_{\text{obs}}|\big)}{B + 1}. \tag{12}$$

Ties are included. The plus-one convention prevents Monte Carlo zero values, but does not make this bootstrap approximation finite-sample exact. Non-smooth statistics such as AP, F1, and binned ECE warrant particular caution.

The principal reporting family contains nine tests: the three pairwise comparisons among NoJEPA, DirectReadout, and ForecastAttention, each for Macro AUROC, AP, and F1. The secondary family contains 81 tests: 27 landmark-specific AUROC/AP/F1 comparisons among the principal models; 33 comparisons of the other 11 Macro metrics; 12 ForecastAttention–ForecastMean comparisons for AUROC/AP/F1 at T0, T1, T2, and Macro; and nine T2–T0 AUROC/AP/F1 comparisons within the three principal models. Holm adjustment (Holm, 1979) is applied separately within each family. The aggregate CSV additionally reports an all-90 Holm sensitivity analysis; separate-family adjustments do not control the familywise error rate over the union. We use $p_{\text{Holm}} < 0.05$, with $*$, $**$, and $***$ denoting adjusted values below 0.05, 0.01, and 0.001, respectively, and ns otherwise.

The 2,000-resample percentile intervals remain unadjusted and are not exact inversions of these tests. We compute p values from the bootstrap differences rather than deriving them from whether an interval includes zero. Seed 44, the metric-averaging policy, testing procedure, and reporting families were chosen after results were available. All inference is exploratory and conditions on these fitted models; it does not measure uncertainty from retraining or retrospective selection. Validation-threshold, subgroup, decision-curve, and training-probe summaries receive no additional significance tests.

5.6. Operating-point and hypothetical decision analyses

As a threshold sensitivity analysis, we select a separate Youden threshold for each variant, fold, and landmark using that fold model's validation predictions. The highest threshold resolves ties. Validation coverage per fold is 160–161 patients at T0, 151–154 at T1, and 138–143 at T2. We then apply the threshold to holdout predictions from that same fold model. We do not combine validation scores across folds, and no holdout outcome selects a threshold. This keeps the validation and test score definitions tied to one fitted model. Validation patients also informed that model's best-checkpoint selection, however, so this is not nested or independent calibration. The resulting holdout metrics are averaged across folds using the same procedure as the fixed-threshold metrics. Patient-bootstrap intervals keep the validation-selected thresholds fixed and exclude their selection uncertainty.

We additionally compute exploratory decision curves (Vickers and Elkin, 2006). At risk threshold q, the generic positive action is applied when predicted pCR probability is at least q. Its mathematical net benefit is

$$\text{NB}(q) = \frac{\text{TP}(q)}{n} - \frac{\text{FP}(q)}{n}\frac{q}{1-q}. \tag{13}$$

We calculate this separately from each fold model's probabilities over thresholds 0.05–0.80, average the three landmark net benefits for Macro within each fold, and then average across the five folds. Treat-all and treat-none are references. The calculation uses uncalibrated probabilities and does not validate a particular escalation, de-escalation, or other management action.

5.7. Exploratory subgroup and mechanism analyses

We examine released HR/HER2 combinations, age below or at least 50 years, native source namespace, and saved predicted-component crop coverage. Age 50 is an exploratory reporting cutpoint. Subgroup AUROC/AP are suppressed when fewer than ten patients or fewer than five outcomes in either class are available. Subgroup metrics are computed separately for each fold model and then equally averaged. Paired within-subgroup patient resampling preserves fold, model, and landmark dependence; single-class draws are omitted for AUROC intervals and valid-draw counts retained. Intervals are unadjusted, and no subgroup-interaction hypothesis is tested. Source namespaces share the same clinical release and are not independent external cohorts.

Mechanism analysis is restricted to archived fixed-training-probe diagnostics collected every five epochs. These probes sample up to 32 training patients and are not full validation or holdout forecast evaluations. Unlike the missing-target-masked training objective, the archived forecast diagnostic includes only continuous observed visit chains: the real landmark, target, and every intermediate canonical visit must be available. A gap excludes that landmark–target pair, and eligible-patient counts therefore vary by horizon. We report a descriptive common epoch 20 and common-epoch trajectories. Forecast skill is $1 - \text{L1}_{\text{forecast}}/\text{L1}_{\text{copy}}$ within the same snapshot's EMA

representation; the copy forecast holds the observed landmark state constant. This normalization avoids comparing raw latent errors between separately learned teacher coordinates. Predictor gradient norms from the pCR and weighted rollout objectives are descriptive checks of gradient routing, not independent statistical samples or clinical effects. These archived diagnostics are computed within each seed-44 fold model's own EMA coordinates and summarized descriptively across the five folds. Only epochs with complete diagnostic coverage are used for common-epoch summaries. Latent states are not combined between models. These training diagnostics are distinct from the fold-specific holdout pCR evaluation.

5.8. Scope of the selected-seed fold-mean analysis

Seed 44 and the decision to report a mean of fold-specific metrics were selected retrospectively after holdout results were available. Applying the same seed, folds, and averaging rule to every model preserves a matched architectural comparison, but does not remove selection bias. The fold standard deviation describes five fitted models trained on overlapping partitions under one seed; it does not characterize training-seed robustness. Patient-bootstrap inference conditions on those fixed models. A confirmatory study should lock training seeds, aggregation, model selection, statistical comparisons, and operating-point policies before evaluating independent patients.

6. Results

6.1. Discrimination of separately evaluated fold models

All 15 principal-model runs and five supplementary mean-pooling runs completed training and formal evaluation. Twenty evaluation receipts and 60 registered report identities were verified. Each seed-44 fold model was evaluated separately on the same 175 holdout patients, including 56 positive outcomes. Table 2 reports the mean and sample standard deviation of the five fold-specific metrics. Macro is first computed within each fold and then averaged across folds; no model probabilities are averaged.

ForecastAttention had fold-mean T0/T1/T2 AUROCs of 0.7294/0.7381/0.7421 and AP values of 0.5924/0.6012/0.6033 (Fig. 2). Its Macro AUROC/AP/F1 was 0.7365/0.5990/0.5469, with fold standard deviations of 0.0159/0.0151/0.0334. Corresponding mean Macro values were 0.7219/0.5792/0.5437 for NoJEPA and 0.7303/0.5888/0.5539 for DirectReadout. Forecast attention had the highest mean Macro AUROC and AP, while direct readout had the highest mean Macro F1 among all four variants. At T0, forecast attention's mean AUROC was slightly below direct readout (0.7294 versus 0.7304), whereas it was higher at T1 and T2. The discrimination ordering was therefore not uniform across landmarks.

Under direct readout, adding future-state supervision increased mean Macro AUROC/AP/F1 by +0.0084/+0.0097/+0.0102. The full forecast-attention model gave further mean AUROC and AP increases of +0.0062 and +0.0101 relative to DirectReadout, but mean F1 decreased by 0.0070. The first comparison more closely isolates auxiliary supervision, whereas the second also changes forecast inputs, pooling, and classifier width. Direct readout had higher Macro AUROC than NoJEPA in all five folds. Forecast attention exceeded both direct controls in four folds but was lower in fold 5: its Macro AUROC was 0.7170, compared with 0.7347 for NoJEPA and 0.7477 for DirectReadout. All fold-specific results are retained in Tables A.7 and A.8.

The supplementary ForecastMean had mean Macro AUROC/AP/F1 of 0.7348/0.5845/0.5458 (Table A.6). Forecast attention's corresponding differences were +0.0017/+0.0144/+0.0011, but all paired intervals included zero. Mean pooling had slightly higher T0 AUROC than forecast attention (0.7317 versus 0.7294). The mean estimates do not establish a consistent attention benefit across folds, landmarks, and metrics.

The paired patient intervals for every principal Macro AUROC/AP/F1 comparison included zero (Fig. 3). The full-versus-NoJEPA AUROC difference was +0.0146 [−0.0073, 0.0376], the DirectReadout–NoJEPA difference +0.0084 [−0.0049, 0.0212], and the full-versus-direct difference +0.0062 [−0.0136, 0.0275]. The full-versus-direct F1 difference was −0.0070 [−0.0398, 0.0242]. These conditional patient intervals are distinct from descriptive fold standard deviations and exclude retraining and seed-selection uncertainty. None of the nine principal Macro comparisons met the 0.05 criterion, and all nine had within-family Holm-adjusted $p = 1.0000$ (Table 3). For example, the full-versus-NoJEPA Macro AUROC difference had raw $p = 0.19246$, and the full-versus-direct comparison had raw $p = 0.54536$. These results do not establish superiority or equivalence; they qualify the descriptive ranking differences conditional on the selected seed and fitted models.

6.2. Threshold behavior and class imbalance

At threshold 0.5, mean Macro sensitivity/NPV was 0.7036/0.8269 for ForecastAttention, compared with 0.6357/0.8023 for DirectReadout. Higher sensitivity accompanied lower specificity (0.5938 versus 0.6908) and PPV (0.4643 versus 0.4938). The paired sensitivity difference was +0.0679 [0.0262, 0.1103], while the specificity difference was −0.0969 [−0.1280, −0.0685]. These operating-point differences should accompany the ranking means. The full-versus-direct specificity decrease met both the secondary-family and all-90 Holm criteria (adjusted $p = 0.00081$ and 0.00090). Relative to NoJEPA, forecast attention had a mean sensitivity increase of +0.0940 and specificity decrease of 0.1109; their secondary-family adjusted values were 0.03002 and 0.00081, respectively, and their all-90 values were 0.03344 and 0.00090. These three operating-point comparisons were the only secondary tests meeting the family-adjusted criterion. The two specificity tests reached the raw Monte Carlo resolution of 10^{-5}, rather than a zero p value.

Table 2: Performance on the same fixed internal holdout (175 patients, 56 pCR positive). Entries are mean ± sample SD of metrics computed separately for the five seed44 fold models. Probabilities are not averaged. Macro first averages T0–T2 metrics within each fold. F1 uses threshold 0.5. Fold SD is descriptive; the five models share the holdout and overlapping training sets. Paired Macro differences are in Table 3.

Model	Landmark	AUROC	AP	F1
NoJEPA	T0	0.7209 ± 0.0117	0.5749 ± 0.0295	0.5441 ± 0.0378
NoJEPA	T1	0.7219 ± 0.0113	0.5815 ± 0.0185	0.5458 ± 0.0408
NoJEPA	T2	0.7230 ± 0.0112	0.5812 ± 0.0145	0.5411 ± 0.0409
NoJEPA	Macro	0.7219 ± 0.0113	0.5792 ± 0.0202	0.5437 ± 0.0383
DirectReadout	T0	0.7304 ± 0.0135	0.5854 ± 0.0200	0.5447 ± 0.0302
DirectReadout	T1	0.7306 ± 0.0120	0.5891 ± 0.0226	0.5485 ± 0.0178
DirectReadout	T2	0.7299 ± 0.0116	0.5920 ± 0.0228	0.5685 ± 0.0329
DirectReadout	Macro	0.7303 ± 0.0124	0.5888 ± 0.0214	0.5539 ± 0.0233
ForecastAttention	T0	0.7294 ± 0.0171	0.5924 ± 0.0124	0.5334 ± 0.0350
ForecastAttention	T1	0.7381 ± 0.0161	0.6012 ± 0.0156	0.5563 ± 0.0330
ForecastAttention	T2	0.7421 ± 0.0167	0.6033 ± 0.0191	0.5510 ± 0.0427
ForecastAttention	Macro	0.7365 ± 0.0159	0.5990 ± 0.0151	0.5469 ± 0.0334

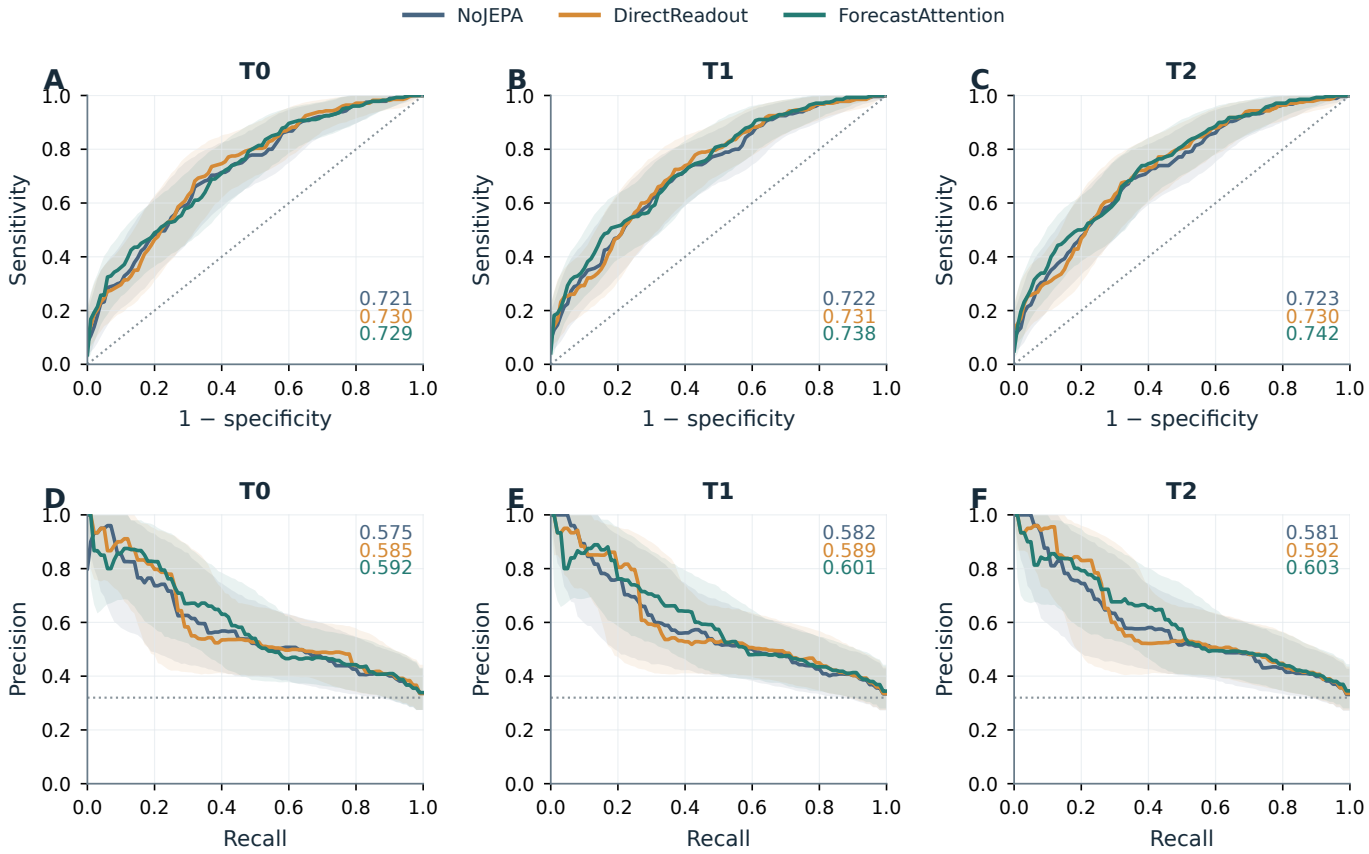


Figure 2: Discrimination of seed-44 fold models evaluated separately. (A–C) Mean ROC curves; (D–F) mean precision–recall curves. Curves are computed per fold, interpolated onto a common grid, and averaged across five folds. Shading gives conditional 95% shared-patient-bootstrap intervals for the mean curve. Colored values summarize directly calculated per-fold AUROC or non-interpolated AP, whose means need not equal the area under the displayed mean curve. The dotted PR reference is the 0.32 positive prevalence. No probabilities are averaged between fold models.

The full model's fold-mean true-positive counts at T0/T1/T2 were 36.2/42.0/40.0, with false-positive counts 42.6/52.8/49.6. These are averages of integer counts from five correlated predictors, not fractional patients or a combined confusion matrix. At T1, sensitivity rose to 0.75 while specificity fell to 0.5563. Its F1 was 0.5334/0.5563/0.5510 across T0/T1/T2, exceeding direct readout only at T1. Improved ranking and a fixed-threshold decision rule therefore gave different patterns.

Among the principal variants, DirectReadout had the highest mean Macro balanced accuracy (0.6632) and MCC (0.3109), compared with 0.6571/0.3000 for NoJEPA and

Table 3: Exploratory principal Macro comparisons. All metrics use approximate, null-centered paired patient-bootstrap tests with 99,999 shared resamples. Within each draw, metrics are computed for each fold model then equally averaged; probabilities are never averaged. All p values are two-sided. Holm$_9$ adjusts the nine entries here; Holm$_{90}$ is the sensitivity analysis over all principal and secondary tests. Sig. refers to Holm$_9$. Direct and Attention abbreviate DirectReadout and ForecastAttention. $^{*}p_{\text{Holm}} < 0.05$, $^{**}p_{\text{Holm}} < 0.01$, $^{***}p_{\text{Holm}} < 0.001$; ns, not significant after the stated correction. Intervals are unadjusted 95% patient-bootstrap intervals (2,000 shared draws) and need not agree with adjusted test decisions. All analyses condition on the five fixed seed44 fold models; folds are not resampled and share one holdout.

Comparison	Metric	Difference [95% CI]	p	$p_{\text{Holm},9}$	$p_{\text{Holm},90}$	Sig.
Direct – NoJEPA	AUROC	+0.0084 [-0.0049, +0.0212]	0.2199	1.0000	1.0000	ns
Direct – NoJEPA	AP	+0.0097 [-0.0129, +0.0297]	0.3731	1.0000	1.0000	ns
Direct – NoJEPA	F1	+0.0102 [-0.0118, +0.0323]	0.3595	1.0000	1.0000	ns
Attention – NoJEPA	AUROC	+0.0146 [-0.0073, +0.0376]	0.1925	1.0000	1.0000	ns
Attention – NoJEPA	AP	+0.0198 [-0.0158, +0.0621]	0.3060	1.0000	1.0000	ns
Attention – NoJEPA	F1	+0.0032 [-0.0339, +0.0379]	0.8616	1.0000	1.0000	ns
Attention – Direct	AUROC	+0.0062 [-0.0136, +0.0275]	0.5454	1.0000	1.0000	ns
Attention – Direct	AP	+0.0101 [-0.0239, +0.0518]	0.5816	1.0000	1.0000	ns
Attention – Direct	F1	-0.0070 [-0.0398, +0.0242]	0.6600	1.0000	1.0000	ns

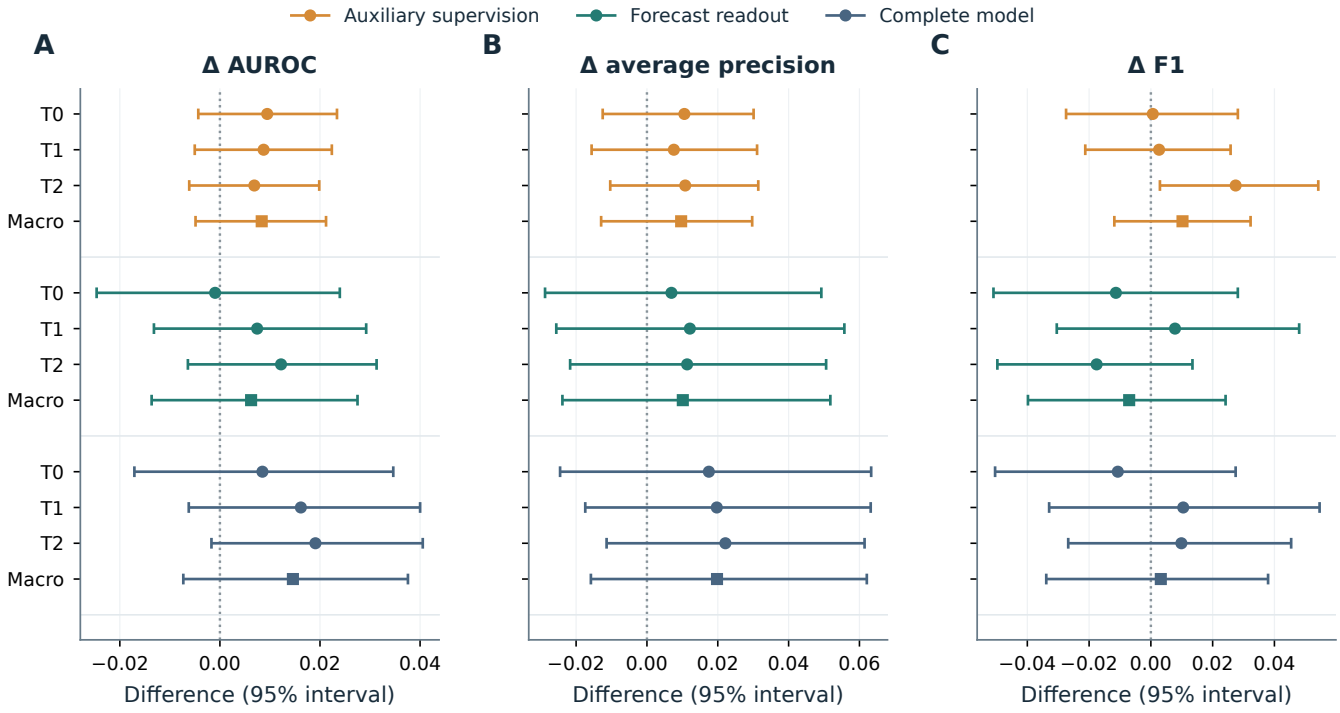


Figure 3: Paired differences in fold-mean metrics at T0, T1, T2, and Macro. Colors distinguish auxiliary supervision, additional forecast readout, and the complete-model contrast. Squares denote Macro estimates. Bars are unadjusted 95% patient-bootstrap intervals, with every patient draw shared across folds, models, and landmarks. Metrics are computed separately within folds before averaging. Adjusted tests appear in Tables 3 and A.11. The five fold models do not create independent test cohorts.

0.6487/0.2931 for ForecastAttention. Mean Macro accuracy was 0.6731, 0.6743, and 0.6290, respectively. All were below the all-negative accuracy reference of 0.68, although direct readout's T2 accuracy was 0.6823. The supplementary ForecastMean had Macro balanced accuracy 0.6529, MCC 0.2990, and accuracy 0.6450. Accuracy alone would obscure the minority-class behavior.

Fold-specific validation Youden thresholds changed mean holdout Macro F1 from 0.5437 to 0.5515 for NoJEPA, 0.5539 to 0.5372 for DirectReadout, 0.5469 to 0.5411 for ForecastAttention, and 0.5458 to 0.5236 for ForecastMean (Table A.10). For forecast attention, mean sensitivity decreased to 0.6440 while specificity increased to 0.6560. Thus the validation-selected thresholds did not consistently improve F1. The thresholds were fitted separately on the matching model's validation predictions and then applied to that model's test predictions; no test outcome selected a threshold. Reuse of validation patients for checkpoint selection limits this analysis to a sensitivity check.

Table 4: Operating characteristics at threshold 0.5. Entries average metrics computed separately for five seed44 fold models; Macro first averages T0–T2 within each fold. No probabilities are averaged. The same 175 patients appear in every evaluation.

Model	Landmark	Sens.	Spec.	PPV	NPV	BA	MCC
NoJEPA	T0	0.614	0.701	0.491	0.796	0.658	0.301
NoJEPA	T1	0.614	0.703	0.492	0.796	0.658	0.302
NoJEPA	T2	0.600	0.711	0.494	0.791	0.655	0.298
NoJEPA	Macro	0.610	0.705	0.492	0.794	0.657	0.300
DirectReadout	T0	0.625	0.686	0.485	0.796	0.655	0.296
DirectReadout	T1	0.625	0.692	0.492	0.798	0.659	0.303
DirectReadout	T2	0.657	0.694	0.504	0.813	0.676	0.334
DirectReadout	Macro	0.636	0.691	0.494	0.802	0.663	0.311
ForecastAttention	T0	0.646	0.642	0.473	0.809	0.644	0.285
ForecastAttention	T1	0.750	0.556	0.455	0.836	0.653	0.298
ForecastAttention	T2	0.714	0.583	0.465	0.836	0.649	0.297
ForecastAttention	Macro	0.704	0.594	0.464	0.827	0.649	0.293

6.3. Longitudinal information and probability quality

The paired fold-mean T2–T0 AUROC change was +0.0127 [0.0026, 0.0235] for ForecastAttention, +0.0021 [−0.0027, 0.0073] for NoJEPA, and −0.0005 [−0.0086, 0.0081] for DirectReadout. Forecast attention's corresponding AP change was +0.0110 [−0.0048, 0.0264] and F1 change +0.0176 [−0.0051, 0.0424]. Its AUROC interval excluded zero without multiplicity correction, whereas the AP and F1 intervals did not. The AUROC change had raw $p = 0.01892$ but secondary-family and all-90 Holm $p = 1.0000$; it therefore did not meet the adjusted criterion. Outcome-conditioned probability-change distributions in Fig. A.8 retain the separate fold predictions.

Mean Macro NLL/Brier/ECE was 0.6149/0.2147/0.1622 for ForecastAttention, compared with 0.5995/0.2073/0.1377 for DirectReadout and 0.5974/0.2069/0.1310 for NoJEPA. Forecast attention had worse point estimates of all three probability-quality metrics than either direct control. The supplementary mean-pooling control had NLL/Brier/ECE of 0.6174/0.2115/0.1524. A higher mean AUROC therefore did not imply better probability quality. All 14 Macro metrics and conditional patient intervals appear in Table A.9.

6.4. Hypothetical net benefit

At a hypothetical threshold of 0.30, mean Macro net benefit was 0.1100 for DirectReadout and 0.0850 for ForecastAttention. At 0.50, all principal variants had negative mean Macro net benefit: −0.0057 for NoJEPA, −0.0069 for DirectReadout, and −0.0510 for ForecastAttention. At 0.70, forecast attention had +0.0013, compared with −0.0801 for direct readout. The ordering therefore depended on threshold. These fold means use uncalibrated probabilities and a generic positive-action utility; they do not establish a clinical threshold or validate a management pathway.

6.5. Exploratory biological and acquisition-related subgroups

The full-versus-NoJEPA mean Macro AUROC differences varied across HR/HER2 groups (Fig. 5; Table B.15). They were +0.0324 [−0.0153, 0.0811] in HR-positive/HER2-negative patients (68 patients, 11 positive), −0.0415 [−0.1353, 0.0457] in HR-positive/HER2-positive patients (31, 12 positive), +0.0082 [−0.1483, 0.1483] in HR-negative/HER2-positive patients (14, 7 positive), and +0.0217 [−0.0245, 0.0667] in HR-negative/HER2-negative patients (62, 26 positive). Every interval included zero, and one subgroup's point estimate favored NoJEPA. These unadjusted exploratory summaries do not establish a subtype-specific benefit or interaction.

In HR-positive/HER2-negative patients, forecast attention's mean Macro sensitivity/specificity was 0.2970/0.8713, compared with 0.0970/0.9696 for NoJEPA and 0.1576/0.9591 for DirectReadout. Greater sensitivity thus accompanied lower specificity in this group. Age and source-namespace summaries are descriptive; the namespaces share one clinical release and do not constitute external validation. The partially clipped baseline-component group contained 11 patients and only one positive outcome, so its AUROC and AP were suppressed under the minimum-outcome rule.

6.6. Archived forecast and gradient diagnostics

Archived fixed-training-probe diagnostics describe individual fold models separately from the pCR holdout analysis (Fig. 6). At common epoch 20, mean copy-normalized forecast skills for DirectReadout were +0.0258, +0.0500, and +0.1935 at one-, two-, and three-step horizons. Corresponding values for ForecastAttention were −0.0368, −0.0154, and +0.0242 (Table B.16). Statistics were normalized within each snapshot's EMA coordinates and averaged across the five seed-44 fold models. Latent states were not combined. These diagnostics do not support attributing forecast attention's ranking means to uniformly more accurate latent forecasts.

Table 5: Probability quality: five-fold metric mean ± sample fold SD for seed44. Metrics are computed separately from each model score vector; Macro averages T0–T2 within fold. Lower is better; ECE uses ten equal-width bins.

Model	NLL	Brier	ECE
NoJEPA	0.5974 ± 0.0111	0.2069 ± 0.0045	0.1310 ± 0.0119
DirectReadout	0.5995 ± 0.0228	0.2073 ± 0.0101	0.1377 ± 0.0191
ForecastAttention	0.6149 ± 0.0553	0.2147 ± 0.0252	0.1622 ± 0.0604

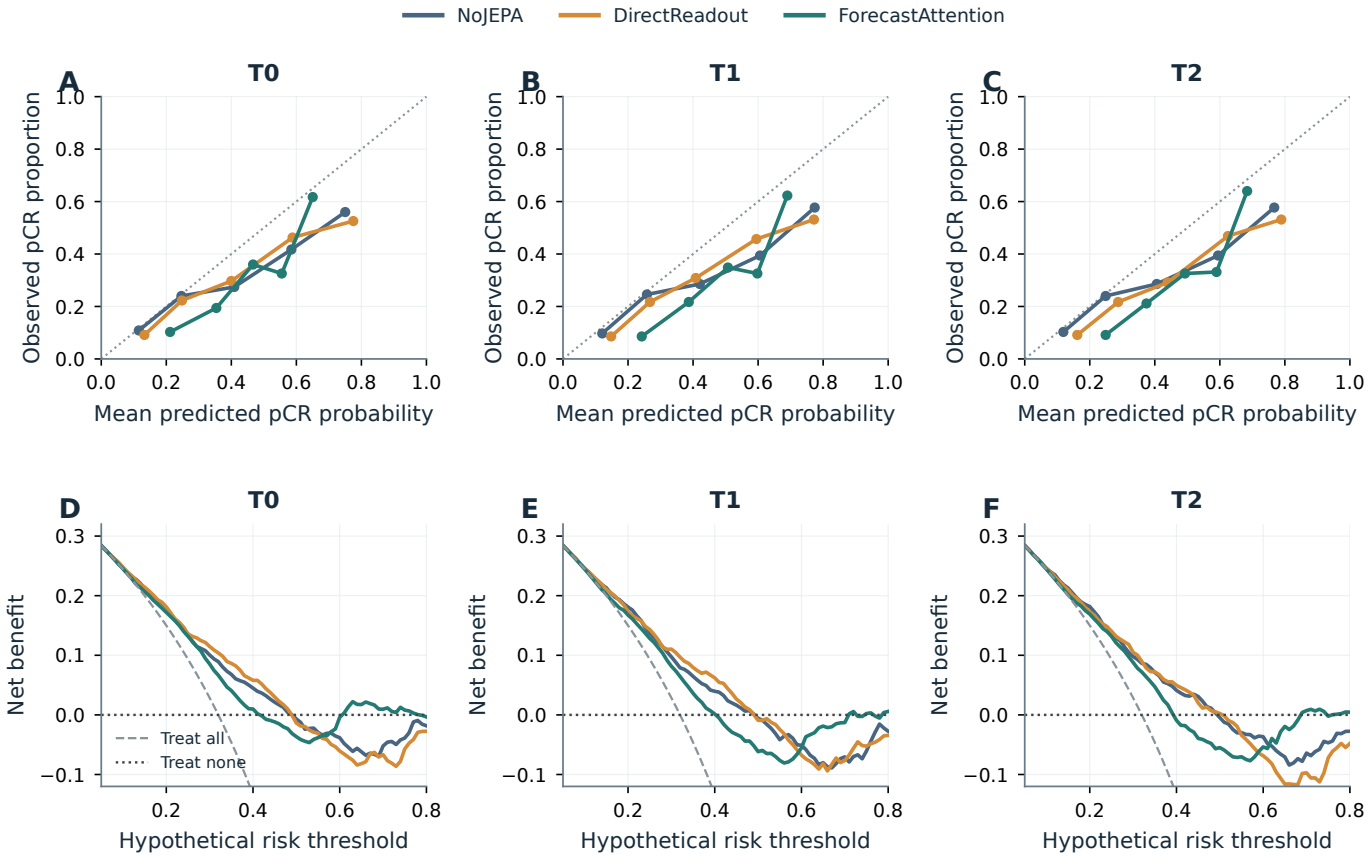


Figure 4: Probability quality and hypothetical decision utility of separately evaluated fold models. (A–C) Reliability summaries form five equal-count probability-rank groups of 35 patients within each fold and then average the group summaries across folds; these differ from the ten equal-width bins used for ECE. (D–F) Net-benefit curves are computed per fold and averaged, with treat-all and treat-none references. Scores are not recalibrated or averaged between models. No clinical escalation or de-escalation strategy is validated.

The probe requires the real landmark, target, and every intermediate examination, whereas the training objective masks unavailable future targets. Eligible populations therefore vary by horizon; only 5–9 patients per probe were eligible at three steps. These are not selected-checkpoint holdout forecast measurements. The predictor's pCR-loss gradient norm was zero for DirectReadout and had mean 0.7148 for ForecastAttention; weighted rollout-gradient means were 0.1213 and 0.1308. These values support the expected gradient routing, but gradient magnitude is not a biological effect size. NoJEPA had no available trained-world diagnostic and was not assigned an artificial zero skill.

7. Discussion

7.1. Separating auxiliary supervision from forecast readout

The mean of the five separately evaluated seed-44 fold models gives a descriptive hierarchy in Macro AUROC and AP: direct future-state supervision improves the direct-readout means, and forecast-attention readout increases them further. Direct readout nevertheless has the highest mean Macro F1. Forecast attention does not lead AUROC at T0 and has lower Macro AUROC than both direct controls in fold 5. None of the nine principal Macro comparisons meets the adjusted criterion. The results therefore describe the selected fitted models without establishing architectural superiority or equivalence.

The supplementary mean-pooling comparison also remains uncertain. Attention has higher mean Macro AU-

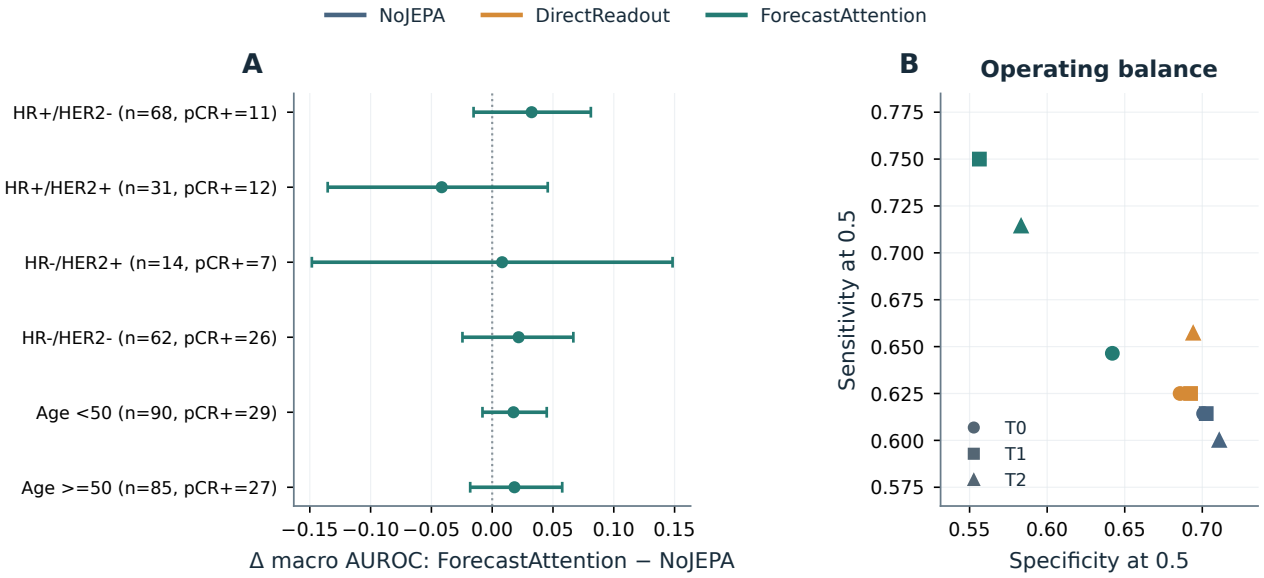


Figure 5: Exploratory subgroup and operating-point summaries, computed separately within seed-44 folds and averaged across folds. (A) Macro AUROC differences between ForecastAttention and NoJEPA within HR/HER2 and age groups, with patient counts, positive outcomes, and unadjusted shared-patient intervals. No subgroup-interaction claim is made. (B) Mean specificity and sensitivity at threshold 0.5 across landmarks; colors identify variants and markers identify landmarks.

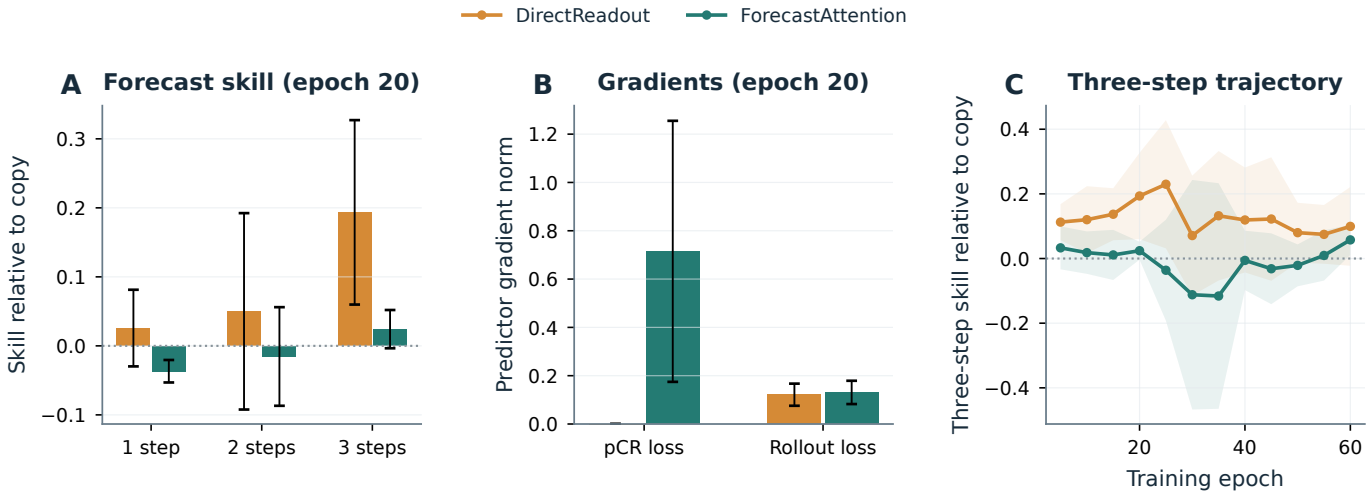


Figure 6: Archived seed-44 fixed-training-probe diagnostics. (A) Copy-normalized forecast skill at common epoch 20. (B) Predictor gradient norms from pCR and weighted rollout losses. (C) Three-step skill across epochs with complete diagnostic coverage. Statistics are calculated within each snapshot's EMA coordinates and summarized descriptively across five folds. These training diagnostics are distinct from holdout pCR metrics; real continuous visit chains are required, and three-step eligibility is only 5–9 patients per probe.

ROC, AP, and F1, but their paired intervals include zero, and the landmark ordering is not uniform. This does not establish that attention is necessary. The DirectReadout–NoJEPA contrast most closely isolates future-state supervision; the full-versus-direct contrast changes forecast features, pooling, and classifier width together. Classifier-width-matched direct readout, stopped-gradient forecasts, copy-state readout, and independent holdout forecast evaluation would help distinguish these contributions.

The main models predict embeddings rather than generate MRI for classification. The internal name NoJEPA denotes removal of the future-state objective while retaining the shared fusion network and transition queries. Archived epoch-20 probes show higher copy-normalized forecast skill for auxiliary-supervised direct models than for forecast attention at every horizon, despite the latter's higher mean pCR AUROC. A possible mechanism is that endpoint gradients shape forecasts toward classification-relevant directions

at the expense of some EMA-target fidelity. Gradient routing is consistent with that possibility, but training probes in separately learned spaces do not establish it or demonstrate anatomical forecast quality.

7.2. What additional visits change

Forecast attention's mean AUROC increases across T0–T2, with a positive unadjusted T2–T0 interval. The corresponding test does not pass secondary-family or all-90 correction, and the AP and F1 intervals include zero. The T1 operating point increases sensitivity while adding false positives and reducing specificity. Added visits, ranking, and a fixed-threshold decision rule therefore need not improve together. Outcome-conditioned histograms summarize within-fold probability changes and do not establish improved predictions for every patient or individual treatment effects.

T0 imaging preceded randomization, while the model is conditioned on the eventual assigned arm. This remains a retrospective conditional scenario. The arm encoding does not capture delivered dose, cycle timing, dose reductions, adherence, or treatment changes. Altering the arm token cannot identify an individual treatment effect without suitable causal assumptions and exposure data. Complete T0–T2 holdout eligibility also restricts the population to which the longitudinal comparison applies.

7.3. Classification, probability quality, and hypothetical utility

The additional metrics reveal operating trade-offs that are obscured by AUROC. Forecast attention has higher mean sensitivity and NPV but lower specificity and PPV than auxiliary-supervised direct readout. Relative to NoJEPA, both the sensitivity increase and specificity decrease meet the secondary-family and all-90 adjusted criteria. Its specificity decrease relative to direct readout also meets both corrections. These are conditional findings about fixed fitted models at threshold 0.5, and the adverse specificity result should accompany the sensitivity gain. All principal models' Macro raw accuracies are below the all-negative reference, and direct readout has the highest mean balanced accuracy and MCC among them.

Forecast attention also has worse NLL, Brier score, and ECE point estimates than both direct controls. Discrimination ordering therefore does not establish probability calibration. Selecting a Youden threshold from each matching fold validation set increases mean F1 only for NoJEPA and decreases it for the other models. These thresholds avoid combining incompatible score definitions across models, but the validation patients also informed checkpoint selection, preventing an independent or nested calibration claim.

Net-benefit orderings change with the hypothetical threshold. Direct readout has higher mean Macro net benefit at 0.30 and 0.50, while forecast attention is higher at 0.70; every principal variant has negative mean Macro net benefit at 0.50. These are fold means under a generic positive-action utility, not evaluations of a defined care pathway. A clinical threshold requires the consequences of false-positive and false-negative actions to be specified and calibration to be assessed without holdout leakage.

7.4. Fold variation and subgroup interpretation

The mean and sample standard deviation across folds describe five models fitted to overlapping development partitions under one seed. Because these models reuse the same holdout, fold standard deviations do not supply independent clinical replications, training-seed variability, or patient standard errors. Shared-patient-bootstrap inference instead measures conditional patient-sampling uncertainty for the average metric of these fixed models. Folds are not resampled and models are not refitted, so those intervals do not characterize the whole training procedure.

HR/HER2 subgroup intervals all include zero; the HR-positive/HER2-positive point estimate favors NoJEPA, whereas the other three groups favor forecast attention descriptively. No subgroup-interaction hypothesis is tested, and the smallest group contains only 14 patients. Source namespaces share one clinical release and do not supply external validation. The crop-clipping subgroup contains 11 patients and one positive outcome; suppressing its AUROC/AP is more informative than interpreting an unstable discrimination estimate.

7.5. Limitations and a confirmatory study

Seed 44, separate-fold metric averaging, and the reporting procedures were selected after holdout results were available. Patient-bootstrap inference conditions on these fitted models and fixed validation-selected thresholds. It excludes retraining, seed choice, model selection, aggregation choice, and threshold-selection uncertainty. One selected seed does not establish training stability, and five overlapping training folds are not a substitute for independent training-seed assessment. The reported mean is the average performance of five distinct predictors, not a deployed combined predictor or a model retrained on the full development cohort.

The 95% patient intervals are unadjusted. Holm corrections within nine principal and 81 secondary comparisons, together with the all-90 sensitivity analysis, do not undo retrospective choices. Null-centered bootstrap tests are approximate, particularly for non-smooth statistics such as AP, F1, and binned ECE. The two specificity tests reached the Monte Carlo resolution floor, which limits precision for their raw p values. Nonsignificant discrimination contrasts do not establish equivalence. A confirmatory study should lock training seeds, aggregation, model and threshold selection, and statistical comparisons before evaluating independent patients.

The holdout is an internal downstream split. Twenty-nine patients had belonged to an earlier development split. Current supervised models were retrained using the new

patient-disjoint folds and were not warm-started from those earlier pCR models, but codec and localization-model pretraining have not been fully audited for current-holdout overlap. This prevents a claim that every pipeline stage was evaluated on wholly unseen patients.

Fixed baseline localization and cropping may omit multifocal disease or changes outside the field of view. Geometry and array checks do not establish anatomical registration accuracy. Holdout patients have complete T0–T2 imaging, whereas development retains incomplete histories; missingness, acquisition variation, and subtype prevalence may limit transportability. Archived fixed-training-probe diagnostics do not replace best-checkpoint holdout forecast evaluation and have small long-horizon denominators.

The study remains an internal architectural comparison. Matched clinical-only, conventional imaging, and independent longitudinal baselines are still needed, along with external validation, anatomical quality review, and independently assessed calibration. Precise pCR pathology criteria, scan intervals, ethics determination, and upstream provenance also require final documentation before submission.

8. Conclusion

Using seed 44 and evaluating five fold models separately, forecast-attention readout had the highest mean Macro AUROC and AP, while auxiliary-supervised direct readout had the highest mean Macro F1. None of the nine principal comparisons met its adjusted significance criterion. Forecast attention's greater sensitivity accompanied lower specificity, and several operating-point contrasts remained detectable after correction. Probability-quality, subgroup, and hypothetical-utility analyses further qualified the discrimination means. These exploratory results motivate independent validation, complete upstream provenance, stronger matched baselines, and prospectively fixed training and statistical policies.

Data and code availability

Source imaging collections are available through The Cancer Imaging Archive, subject to their applicable access conditions and licenses. The present derived manifests, processing lineage, and evaluation outputs are retained locally. [TO COMPLETE: Specify the public repository, versioned release, permitted derived-data artifacts, and access mechanism before submission; no public code-release claim is made in this draft.]

Ethics approval and consent

[TO COMPLETE: Provide the original studies' ethics and consent statements and the local institutional determination for this secondary analysis, including an approval or exemption identifier where applicable. Public availability alone is not an ethics approval.]

Funding

[TO COMPLETE: Provide grant numbers and the funders' roles, or confirm that no specific funding was received.]

Declaration of competing interest

[TO COMPLETE: Obtain and insert the actual declarations from every author.]

CRediT authorship contribution statement

[TO COMPLETE: Provide author names and agreed contributions; no authorship or contributions have been inferred.]

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this initial draft, OpenAI Codex was used to organize source evidence, generate reproducible aggregate tables and figures, and assist with English drafting. [TO COMPLETE: All authors must review and edit the manuscript, verify every statement and reference, and finalize the disclosure under the journal's current policy before submission.]

Appendix A. Supplementary pooling, metrics, and uncertainty

The supplementary ForecastMean uses training seed 44, the same five development folds, observed-history protocol, future-state objective, and forecast feature groups as ForecastAttention, replacing spatial attention with mean pooling. Metrics are calculated separately from each fold model's predictions and then equally averaged. The per-fold results in Tables A.7 and A.8 retain the fitted-model estimates underlying the reported mean and sample standard deviation.

Appendix A.1. Paired significance checks

Table 3 reports the nine principal Macro comparisons, with unadjusted 95% paired patient-bootstrap intervals, raw two-sided approximate bootstrap p values, within-family Holm adjustment, and all-90 Holm adjustment. The secondary family comprises 81 tests in Tables A.11–A.14. Every patient draw is shared across all five fixed fold models, variants, and landmarks. Metrics are recomputed separately for each fold before averaging, so the independent patient count remains 175. A significance marker refers to the stated Holm-adjusted family; ns denotes $p_{\mathrm{Holm}} \geq 0.05$. The 2,000-resample percentile intervals are unadjusted and are not exact inversions of the 99,999-resample null-centered tests. All testing and reporting choices remain retrospective and exploratory.

Appendix B. Supplementary subgroup and training-probe summaries

Appendix C. Reproducibility and reporting items to complete

The source package includes aggregate CSV files, source-report hashes, reproducible metric and figure scripts, the shared-patient-bootstrap procedure, and the official article class. The significance CSV includes raw p values, within-family Holm adjustment, and adjustment over all 90 tests as a sensitivity analysis. It contains no patient identifiers, patient-level prediction rows, original MRI volumes, or model checkpoints. The method figure includes only rendered, anonymous development-image slices. Every variant uses training seed 44. Metrics are computed separately from each fold model's predictions before averaging across folds, with no probability ensembling. Archived training diagnostics are calculated in each model's own latent coordinates and summarized descriptively across folds.

[TO COMPLETE: Before submission, complete author and ethics statements; verify the precise pCR pathology definition; document codec and segmentation patient-level pretraining provenance; add matched external baselines and prospectively locked training and analysis policies; audit external-cohort independence; and confirm the journal's current submission and anonymization requirements.]

Table A.6: Supplementary spatial pooling comparison: mean ± sample fold SD of seed44 metrics. Each of the five fold models is evaluated separately; no probabilities are averaged. ForecastMean and ForecastAttention share forecast readout and future-state supervision.

Model	Landmark	AUROC	AP	F1
ForecastMean	T0	0.7317 ± 0.0094	0.5812 ± 0.0088	0.5446 ± 0.0406
ForecastMean	T1	0.7367 ± 0.0100	0.5869 ± 0.0191	0.5523 ± 0.0312
ForecastMean	T2	0.7361 ± 0.0103	0.5856 ± 0.0189	0.5405 ± 0.0172
ForecastMean	Macro	0.7348 ± 0.0092	0.5845 ± 0.0145	0.5458 ± 0.0225
ForecastAttention	T0	0.7294 ± 0.0171	0.5924 ± 0.0124	0.5334 ± 0.0350
ForecastAttention	T1	0.7381 ± 0.0161	0.6012 ± 0.0156	0.5563 ± 0.0330
ForecastAttention	T2	0.7421 ± 0.0167	0.6033 ± 0.0191	0.5510 ± 0.0427
ForecastAttention	Macro	0.7365 ± 0.0159	0.5990 ± 0.0151	0.5469 ± 0.0334

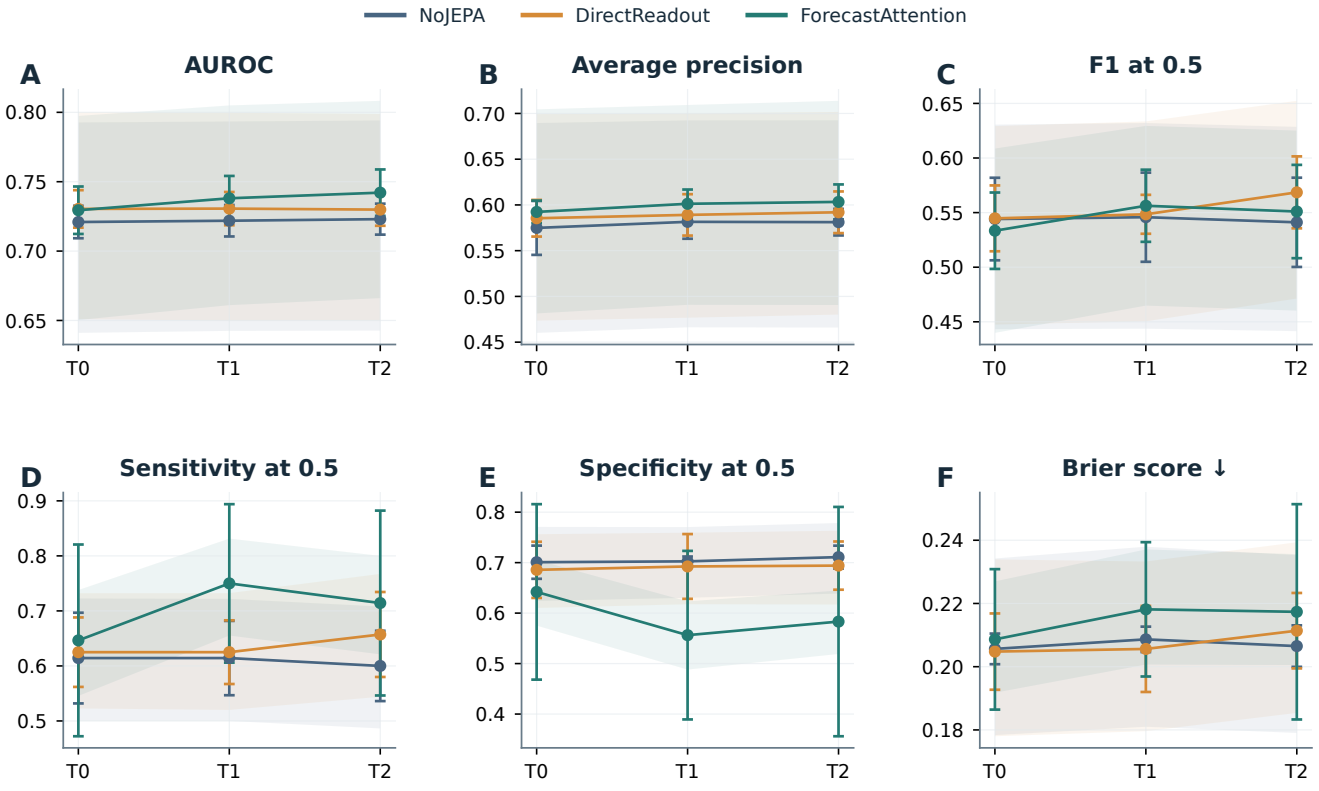


Figure A.7: Six complementary landmark summaries: AUROC, AP, F1, sensitivity, specificity, and Brier score. Metrics are computed separately for each seed-44 fold model and then equally averaged across five folds. Error bars show descriptive sample fold standard deviations, while shading gives conditional 95% shared-patient-bootstrap intervals for the fold mean. All models and landmarks use the same 175 patients.

Table A.7: Individual seed44 fold-model results for direct readouts. Each fold model is evaluated separately on the same 175 patients (56 positive), without averaging probabilities. Macro averages the three landmark metrics. The folds are overlapping training partitions, not independent test cohorts.

Model	Fold	Landmark	AUROC	AP	F1
NoJEPA	1	T0	0.7131	0.5718	0.5254
NoJEPA	1	T1	0.7165	0.5952	0.5469
NoJEPA	1	T2	0.7150	0.5872	0.5299
NoJEPA	1	Macro	0.7149	0.5848	0.5341
NoJEPA	2	T0	0.7339	0.6042	0.5758
NoJEPA	2	T1	0.7332	0.5959	0.6061
NoJEPA	2	T2	0.7342	0.5986	0.6000
NoJEPA	2	Macro	0.7338	0.5996	0.5939
NoJEPA	3	T0	0.7090	0.5267	0.5354
NoJEPA	3	T1	0.7110	0.5506	0.5246
NoJEPA	3	T2	0.7149	0.5588	0.5238
NoJEPA	3	Macro	0.7116	0.5453	0.5279
NoJEPA	4	T0	0.7156	0.5811	0.4957
NoJEPA	4	T1	0.7137	0.5809	0.4959
NoJEPA	4	T2	0.7147	0.5817	0.4918
NoJEPA	4	Macro	0.7147	0.5812	0.4945
NoJEPA	5	T0	0.7330	0.5906	0.5882
NoJEPA	5	T1	0.7350	0.5851	0.5556
NoJEPA	5	T2	0.7362	0.5795	0.5600
NoJEPA	5	Macro	0.7347	0.5851	0.5679
DirectReadout	1	T0	0.7305	0.5843	0.5426
DirectReadout	1	T1	0.7291	0.5796	0.5672
DirectReadout	1	T2	0.7291	0.5790	0.5857
DirectReadout	1	Macro	0.7296	0.5810	0.5652
DirectReadout	2	T0	0.7372	0.6104	0.5424
DirectReadout	2	T1	0.7374	0.6141	0.5500
DirectReadout	2	T2	0.7350	0.6141	0.6087
DirectReadout	2	Macro	0.7365	0.6129	0.5670
DirectReadout	3	T0	0.7150	0.5940	0.5041
DirectReadout	3	T1	0.7179	0.6003	0.5203
DirectReadout	3	T2	0.7183	0.6019	0.5197
DirectReadout	3	Macro	0.7171	0.5987	0.5147
DirectReadout	4	T0	0.7203	0.5554	0.5455
DirectReadout	4	T1	0.7213	0.5552	0.5455
DirectReadout	4	T2	0.7201	0.5586	0.5620
DirectReadout	4	Macro	0.7206	0.5564	0.5510
DirectReadout	5	T0	0.7489	0.5832	0.5891
DirectReadout	5	T1	0.7473	0.5964	0.5593
DirectReadout	5	T2	0.7468	0.6062	0.5667
DirectReadout	5	Macro	0.7477	0.5953	0.5717

Table A.8: Individual seed44 fold-model results for forecast readouts. Evaluation and aggregation match Table A.7; these individual fold metrics generate the mean and sample SD in the main tables.

Model	Fold	Landmark	AUROC	AP	F1
ForecastAttention	1	T0	0.7449	0.6015	0.5802
ForecastAttention	1	T1	0.7410	0.6033	0.6056
ForecastAttention	1	T2	0.7407	0.5973	0.6074
ForecastAttention	1	Macro	0.7422	0.6007	0.5977
ForecastAttention	2	T0	0.7423	0.6024	0.5591
ForecastAttention	2	T1	0.7584	0.6218	0.5514
ForecastAttention	2	T2	0.7688	0.6311	0.5288
ForecastAttention	2	Macro	0.7565	0.6185	0.5464
ForecastAttention	3	T0	0.7368	0.5952	0.5047
ForecastAttention	3	T1	0.7471	0.6010	0.5179
ForecastAttention	3	T2	0.7446	0.6048	0.5470
ForecastAttention	3	Macro	0.7428	0.6003	0.5232
ForecastAttention	4	T0	0.7176	0.5909	0.5000
ForecastAttention	4	T1	0.7237	0.6021	0.5389
ForecastAttention	4	T2	0.7311	0.6054	0.4961
ForecastAttention	4	Macro	0.7241	0.5995	0.5117
ForecastAttention	5	T0	0.7056	0.5719	0.5231
ForecastAttention	5	T1	0.7200	0.5780	0.5676
ForecastAttention	5	T2	0.7254	0.5780	0.5755
ForecastAttention	5	Macro	0.7170	0.5759	0.5554
ForecastMean	1	T0	0.7369	0.5746	0.5600
ForecastMean	1	T1	0.7469	0.5831	0.5882
ForecastMean	1	T2	0.7482	0.5963	0.5556
ForecastMean	1	Macro	0.7440	0.5846	0.5679
ForecastMean	2	T0	0.7458	0.5702	0.6015
ForecastMean	2	T1	0.7458	0.5670	0.5487
ForecastMean	2	T2	0.7458	0.5666	0.5192
ForecastMean	2	Macro	0.7458	0.5679	0.5565
ForecastMean	3	T0	0.7263	0.5922	0.4906
ForecastMean	3	T1	0.7248	0.6119	0.5045
ForecastMean	3	T2	0.7321	0.6038	0.5299
ForecastMean	3	Macro	0.7277	0.6026	0.5083
ForecastMean	4	T0	0.7251	0.5856	0.5399
ForecastMean	4	T1	0.7285	0.6009	0.5506
ForecastMean	4	T2	0.7290	0.5975	0.5600
ForecastMean	4	Macro	0.7275	0.5946	0.5501
ForecastMean	5	T0	0.7242	0.5833	0.5312
ForecastMean	5	T1	0.7377	0.5716	0.5693
ForecastMean	5	T2	0.7252	0.5636	0.5378
ForecastMean	5	Macro	0.7290	0.5728	0.5461

Table A.9: All 14 macro metrics for the principal variants: point estimate [95% patient-bootstrap percentile interval]. Metrics are computed separately for each seed44 fold model, averaged over T0–T2 within fold, then equally averaged over five folds. Probabilities are not averaged. Threshold-dependent metrics use 0.5. Identical patient resamples preserve pairing across models and landmarks. Entries are marginal model intervals; paired differences are reported separately. Intervals condition on the fitted fold models and exclude retraining and retrospective selection uncertainty. Downward arrows identify metrics for which lower is better.

Metric	NoJEPA	DirectReadout	ForecastAttention
AUROC	0.7219 [0.6422, 0.7931]	0.7303 [0.6514, 0.7999]	0.7365 [0.6602, 0.8040]
AP	0.5792 [0.4646, 0.6913]	0.5888 [0.4769, 0.7000]	0.5990 [0.4891, 0.7082]
F1	0.5437 [0.4433, 0.6293]	0.5539 [0.4575, 0.6374]	0.5469 [0.4551, 0.6199]
Accuracy	0.6743 [0.6137, 0.7349]	0.6731 [0.6130, 0.7299]	0.6290 [0.5752, 0.6777]
Sensitivity	0.6095 [0.4930, 0.7183]	0.6357 [0.5321, 0.7436]	0.7036 [0.6125, 0.7877]
Specificity	0.7048 [0.6319, 0.7728]	0.6908 [0.6162, 0.7593]	0.5938 [0.5291, 0.6573]
PPV	0.4921 [0.3813, 0.5991]	0.4938 [0.3874, 0.5960]	0.4643 [0.3679, 0.5562]
NPV	0.7943 [0.7164, 0.8632]	0.8023 [0.7229, 0.8713]	0.8269 [0.7559, 0.8879]
Balanced accuracy	0.6571 [0.5887, 0.7234]	0.6632 [0.5960, 0.7269]	0.6487 [0.5913, 0.7004]
MCC	0.3000 [0.1683, 0.4237]	0.3109 [0.1828, 0.4305]	0.2931 [0.1808, 0.3874]
Youden index	0.3143 [0.1774, 0.4468]	0.3265 [0.1920, 0.4538]	0.2974 [0.1826, 0.4009]
NLL ↓	0.5974 [0.5344, 0.6665]	0.5995 [0.5387, 0.6660]	0.6149 [0.5778, 0.6548]
Brier score ↓	0.2069 [0.1795, 0.2358]	0.2073 [0.1812, 0.2352]	0.2147 [0.1979, 0.2327]
ECE ↓	0.1310 [0.1104, 0.1963]	0.1377 [0.1142, 0.2002]	0.1622 [0.1260, 0.2265]

Table A.10: Threshold sensitivity analysis: seed44 five-fold metric means. Each fold model uses its own landmark-specific validation Youden threshold on the holdout scores from the same model. Validation also informed best-checkpoint selection; this is sensitivity analysis, not nested calibration. No test labels choose thresholds.

Model	Policy	F1	Sens.	Spec.	MCC
NoJEPA	0.5	0.544	0.610	0.705	0.300
NoJEPA	Fold validation Youden	0.551	0.645	0.675	0.303
DirectReadout	0.5	0.554	0.636	0.691	0.311
DirectReadout	Fold validation Youden	0.537	0.581	0.734	0.310
ForecastAttention	0.5	0.547	0.704	0.594	0.293
ForecastAttention	Fold validation Youden	0.541	0.644	0.656	0.288

Table A.11: Exploratory landmark comparisons. All metrics use approximate, null-centered paired patient-bootstrap tests (99,999 shared draws across folds, models, and landmarks). All p values are two-sided. p_{Holm} adjusts jointly over the 81 secondary tests in Tables A.11–A.14. Direct and Attention abbreviate DirectReadout and ForecastAttention. $^{*}p_{\text{Holm}} < 0.05$, $^{**}p_{\text{Holm}} < 0.01$, $^{***}p_{\text{Holm}} < 0.001$; ns, not significant after the stated correction. Intervals are unadjusted 95% patient-bootstrap intervals (2,000 shared draws) and need not agree with adjusted test decisions. All analyses condition on the five fixed seed44 fold models; folds are not resampled and share one holdout.

Comparison	Landmark	Metric	Difference [95% CI]	p	p_{Holm}	Sig.
Direct – NoJEPA	T0	AUROC	+0.0095 [-0.0043, +0.0234]	0.1953	1.0000	ns
Direct – NoJEPA	T0	AP	+0.0106 [-0.0125, +0.0301]	0.3397	1.0000	ns
Direct – NoJEPA	T0	F1	+0.0006 [-0.0275, +0.0282]	0.9656	1.0000	ns
Direct – NoJEPA	T1	AUROC	+0.0087 [-0.0050, +0.0224]	0.2221	1.0000	ns
Direct – NoJEPA	T1	AP	+0.0076 [-0.0156, +0.0311]	0.5254	1.0000	ns
Direct – NoJEPA	T1	F1	+0.0027 [-0.0212, +0.0258]	0.8279	1.0000	ns
Direct – NoJEPA	T2	AUROC	+0.0069 [-0.0061, +0.0198]	0.3084	1.0000	ns
Direct – NoJEPA	T2	AP	+0.0108 [-0.0104, +0.0314]	0.3172	1.0000	ns
Direct – NoJEPA	T2	F1	+0.0274 [+0.0029, +0.0542]	0.0309	1.0000	ns
Attention – NoJEPA	T0	AUROC	+0.0085 [-0.0171, +0.0346]	0.5060	1.0000	ns
Attention – NoJEPA	T0	AP	+0.0175 [-0.0245, +0.0633]	0.4277	1.0000	ns
Attention – NoJEPA	T0	F1	-0.0107 [-0.0504, +0.0274]	0.5972	1.0000	ns
Attention – NoJEPA	T1	AUROC	+0.0162 [-0.0062, +0.0400]	0.1564	1.0000	ns
Attention – NoJEPA	T1	AP	+0.0197 [-0.0174, +0.0632]	0.3221	1.0000	ns
Attention – NoJEPA	T1	F1	+0.0105 [-0.0329, +0.0546]	0.6422	1.0000	ns
Attention – NoJEPA	T2	AUROC	+0.0191 [-0.0017, +0.0405]	0.0710	1.0000	ns
Attention – NoJEPA	T2	AP	+0.0222 [-0.0114, +0.0614]	0.2241	1.0000	ns
Attention – NoJEPA	T2	F1	+0.0099 [-0.0268, +0.0454]	0.5894	1.0000	ns
Attention – Direct	T0	AUROC	-0.0010 [-0.0246, +0.0240]	0.9373	1.0000	ns
Attention – Direct	T0	AP	+0.0069 [-0.0288, +0.0492]	0.7232	1.0000	ns
Attention – Direct	T0	F1	-0.0113 [-0.0510, +0.0282]	0.5821	1.0000	ns
Attention – Direct	T1	AUROC	+0.0074 [-0.0132, +0.0292]	0.4835	1.0000	ns
Attention – Direct	T1	AP	+0.0121 [-0.0256, +0.0557]	0.5427	1.0000	ns
Attention – Direct	T1	F1	+0.0078 [-0.0305, +0.0480]	0.6869	1.0000	ns
Attention – Direct	T2	AUROC	+0.0122 [-0.0064, +0.0313]	0.2146	1.0000	ns
Attention – Direct	T2	AP	+0.0113 [-0.0217, +0.0506]	0.5265	1.0000	ns
Attention – Direct	T2	F1	-0.0176 [-0.0497, +0.0135]	0.2732	1.0000	ns

Table A.12: Exploratory Macro operating and probability-quality comparisons. Differences are first-named minus second-named model; negative Brier, NLL, or ECE differences favor the first model. All metrics use approximate, null-centered paired patient-bootstrap tests (99,999 shared draws across folds, models, and landmarks). All p values are two-sided. p_{Holm} adjusts jointly over the 81 secondary tests in Tables A.11–A.14. Direct and Attention abbreviate DirectReadout and ForecastAttention. $^{*}p_{\text{Holm}} < 0.05$, $^{**}p_{\text{Holm}} < 0.01$, $^{***}p_{\text{Holm}} < 0.001$; ns, not significant after the stated correction. Intervals are unadjusted 95% patient-bootstrap intervals (2,000 shared draws) and need not agree with adjusted test decisions. All analyses condition on the five fixed seed44 fold models; folds are not resampled and share one holdout.

Comparison	Metric	Difference [95% CI]	p	p_{Holm}	Sig.
Direct – NoJEPA	Accuracy	-0.0011 [-0.0187, +0.0152]	0.9142	1.0000	ns
Direct – NoJEPA	Sensitivity	+0.0262 [-0.0011, +0.0560]	0.0820	1.0000	ns
Direct – NoJEPA	Specificity	-0.0140 [-0.0361, +0.0062]	0.1997	1.0000	ns
Direct – NoJEPA	PPV	+0.0017 [-0.0202, +0.0232]	0.8776	1.0000	ns
Direct – NoJEPA	NPV	+0.0080 [-0.0054, +0.0216]	0.2505	1.0000	ns
Direct – NoJEPA	Balanced accuracy	+0.0061 [-0.0119, +0.0241]	0.5082	1.0000	ns
Direct – NoJEPA	MCC	+0.0109 [-0.0233, +0.0453]	0.5391	1.0000	ns
Direct – NoJEPA	Youden index	+0.0122 [-0.0237, +0.0482]	0.5082	1.0000	ns
Direct – NoJEPA	Brier	+0.0003 [-0.0050, +0.0057]	0.9099	1.0000	ns
Direct – NoJEPA	NLL	+0.0021 [-0.0113, +0.0156]	0.7649	1.0000	ns
Direct – NoJEPA	ECE	+0.0067 [-0.0191, +0.0242]	0.5498	1.0000	ns
Attention – NoJEPA	Accuracy	-0.0453 [-0.0781, -0.0149]	0.0044	0.3352	ns
Attention – NoJEPA	Sensitivity	+0.0940 [+0.0483, +0.1422]	0.0004	0.0300	*
Attention – NoJEPA	Specificity	-0.1109 [-0.1459, -0.0779]	< 0.0001	0.0008	***
Attention – NoJEPA	PPV	-0.0278 [-0.0652, +0.0033]	0.1083	1.0000	ns
Attention – NoJEPA	NPV	+0.0326 [+0.0075, +0.0584]	0.0149	1.0000	ns
Attention – NoJEPA	Balanced accuracy	-0.0084 [-0.0387, +0.0194]	0.5721	1.0000	ns
Attention – NoJEPA	MCC	-0.0069 [-0.0648, +0.0463]	0.8099	1.0000	ns
Attention – NoJEPA	Youden index	-0.0169 [-0.0774, +0.0387]	0.5721	1.0000	ns
Attention – NoJEPA	Brier	+0.0078 [-0.0054, +0.0207]	0.2409	1.0000	ns
Attention – NoJEPA	NLL	+0.0176 [-0.0152, +0.0490]	0.2809	1.0000	ns
Attention – NoJEPA	ECE	+0.0312 [-0.0121, +0.0507]	0.0906	1.0000	ns
Attention – Direct	Accuracy	-0.0442 [-0.0724, -0.0175]	0.0014	0.1123	ns
Attention – Direct	Sensitivity	+0.0679 [+0.0262, +0.1103]	0.0021	0.1586	ns
Attention – Direct	Specificity	-0.0969 [-0.1280, -0.0685]	< 0.0001	0.0008	***
Attention – Direct	PPV	-0.0295 [-0.0607, -0.0019]	0.0519	1.0000	ns
Attention – Direct	NPV	+0.0246 [+0.0014, +0.0485]	0.0390	1.0000	ns
Attention – Direct	Balanced accuracy	-0.0145 [-0.0419, +0.0104]	0.2732	1.0000	ns
Attention – Direct	MCC	-0.0178 [-0.0695, +0.0300]	0.4783	1.0000	ns
Attention – Direct	Youden index	-0.0291 [-0.0838, +0.0209]	0.2732	1.0000	ns
Attention – Direct	Brier	+0.0075 [-0.0052, +0.0194]	0.2189	1.0000	ns
Attention – Direct	NLL	+0.0155 [-0.0146, +0.0439]	0.2952	1.0000	ns
Attention – Direct	ECE	+0.0245 [-0.0070, +0.0403]	0.0685	1.0000	ns

Table A.13: Exploratory pooling comparisons: ForecastAttention minus ForecastMean. All metrics use approximate, null-centered paired patient-bootstrap tests (99,999 shared draws across folds, models, and landmarks). All p values are two-sided. p_{Holm} adjusts jointly over the 81 secondary tests in Tables A.11–A.14. Direct and Attention abbreviate DirectReadout and ForecastAttention. $^{*}p_{\text{Holm}} < 0.05$, $^{**}p_{\text{Holm}} < 0.01$, $^{***}p_{\text{Holm}} < 0.001$; ns, not significant after the stated correction. Intervals are unadjusted 95% patient-bootstrap intervals (2,000 shared draws) and need not agree with adjusted test decisions. All analyses condition on the five fixed seed44 fold models; folds are not resampled and share one holdout.

Landmark	Metric	Difference [95% CI]	p	p_{Holm}	Sig.
T0	AUROC	-0.0022 [-0.0234, +0.0191]	0.8273	1.0000	ns
T0	AP	+0.0112 [-0.0194, +0.0465]	0.5022	1.0000	ns
T0	F1	-0.0112 [-0.0516, +0.0289]	0.5810	1.0000	ns
T1	AUROC	+0.0013 [-0.0174, +0.0203]	0.8900	1.0000	ns
T1	AP	+0.0144 [-0.0154, +0.0469]	0.3622	1.0000	ns
T1	F1	+0.0040 [-0.0212, +0.0282]	0.7540	1.0000	ns
T2	AUROC	+0.0060 [-0.0125, +0.0252]	0.5291	1.0000	ns
T2	AP	+0.0178 [-0.0103, +0.0470]	0.2229	1.0000	ns
T2	F1	+0.0105 [-0.0220, +0.0450]	0.5512	1.0000	ns
Macro	AUROC	+0.0017 [-0.0169, +0.0207]	0.8580	1.0000	ns
Macro	AP	+0.0144 [-0.0145, +0.0453]	0.3380	1.0000	ns
Macro	F1	+0.0011 [-0.0237, +0.0271]	0.9351	1.0000	ns

Table A.14: Exploratory paired T2 minus T0 comparisons. Shared patient-bootstrap draws retain dependence between folds and landmarks. All metrics use approximate, null-centered paired patient-bootstrap tests (99,999 shared draws across folds, models, and landmarks). All p values are two-sided. p_{Holm} adjusts jointly over the 81 secondary tests in Tables A.11–A.14. Direct and Attention abbreviate DirectReadout and ForecastAttention. $^{*}p_{\text{Holm}} < 0.05$, $^{**}p_{\text{Holm}} < 0.01$, $^{***}p_{\text{Holm}} < 0.001$; ns, not significant after the stated correction. Intervals are unadjusted 95% patient-bootstrap intervals (2,000 shared draws) and need not agree with adjusted test decisions. All analyses condition on the five fixed seed44 fold models; folds are not resampled and share one holdout.

Model	Metric	Difference [95% CI]	p	p_{Holm}	Sig.
NoJEPA	AUROC	+0.0021 [-0.0027, +0.0073]	0.4258	1.0000	ns
NoJEPA	AP	+0.0063 [-0.0093, +0.0197]	0.4193	1.0000	ns
NoJEPA	F1	-0.0030 [-0.0163, +0.0091]	0.6396	1.0000	ns
DirectReadout	AUROC	-0.0005 [-0.0086, +0.0081]	0.9031	1.0000	ns
DirectReadout	AP	+0.0065 [-0.0062, +0.0189]	0.3131	1.0000	ns
DirectReadout	F1	+0.0238 [+0.0009, +0.0478]	0.0443	1.0000	ns
ForecastAttention	AUROC	+0.0127 [+0.0026, +0.0235]	0.0189	1.0000	ns
ForecastAttention	AP	+0.0110 [-0.0048, +0.0264]	0.1848	1.0000	ns
ForecastAttention	F1	+0.0176 [-0.0051, +0.0424]	0.1341	1.0000	ns

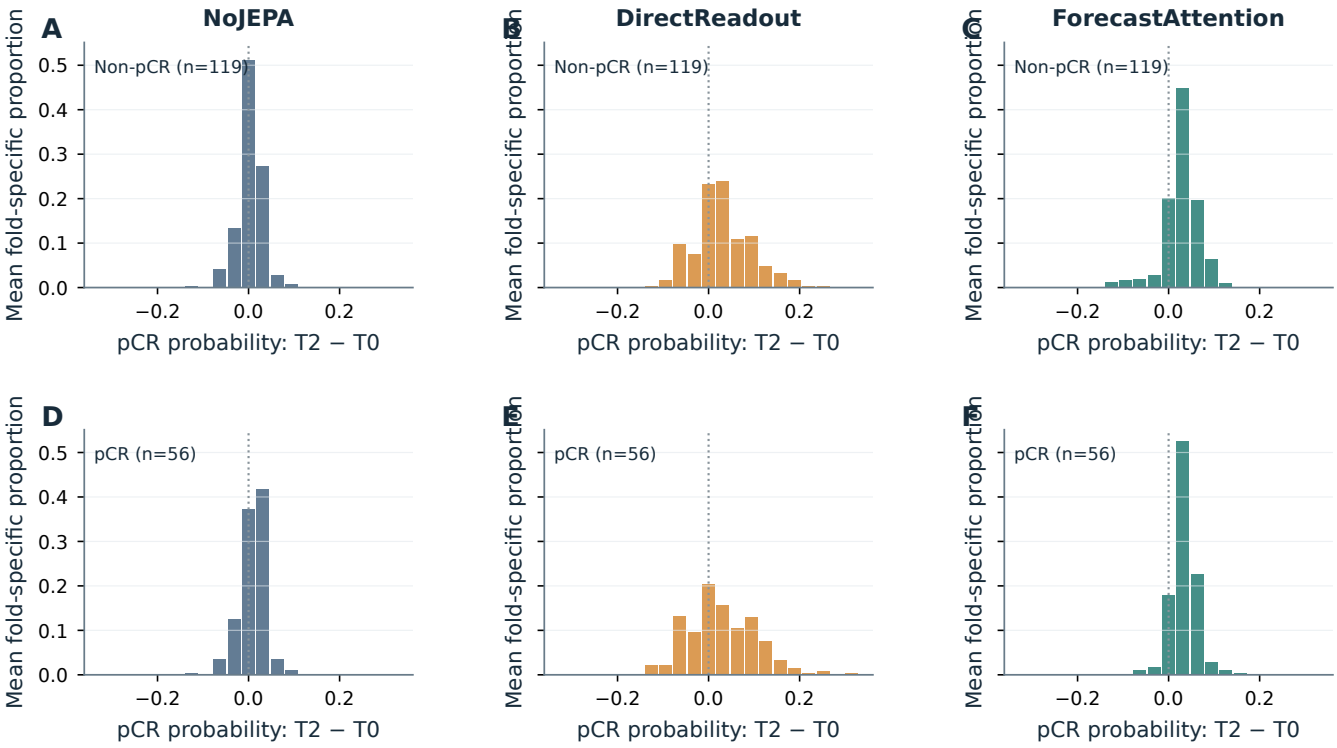


Figure A.8: Outcome-conditioned changes in predicted pCR probability from T0 to T2. Each fold model supplies one probability change per patient. Histograms are constructed separately within folds and then equally averaged; probabilities are not averaged between models. Panels separate variants and outcome classes. Repeated fold predictions are correlated observations of the same 175 patients and do not multiply the sample size by five. Score changes do not establish individual treatment effects.

Table B.15: Exploratory subgroup discrimination using seed 44, with each fold model evaluated separately. Model entries are means ± sample SD of five independently calculated fold metrics; Macro first averages T0–T2 within each fold. No prediction probabilities are averaged. Paired patient-bootstrap intervals for the mean ForecastAttention minus NoJEPA difference condition on the five fixed fold models. Age 50 is an exploratory reporting cutpoint. Namespace groups are parts of one clinical release, not external cohorts.

Subgroup (n/pCR+)	NoJEPA	Full	Δ AUROC [95% interval]
HR+/HER2- (68/11)	0.668 ± 0.032	0.701 ± 0.053	+0.032 [-0.015, +0.081]
HR+/HER2+ (31/12)	0.649 ± 0.022	0.607 ± 0.019	-0.042 [-0.135, +0.046]
HR-/HER2+ (14/7)	0.816 ± 0.075	0.824 ± 0.048	+0.008 [-0.148, +0.148]
HR-/HER2- (62/26)	0.637 ± 0.031	0.658 ± 0.033	+0.022 [-0.024, +0.067]
Age <50 (90/29)	0.667 ± 0.024	0.685 ± 0.028	+0.017 [-0.008, +0.045]
Age ≥50 (85/27)	0.781 ± 0.017	0.800 ± 0.009	+0.018 [-0.018, +0.058]
ISPY2 namespace (125/39)	0.679 ± 0.022	0.691 ± 0.011	+0.012 [-0.016, +0.042]
ACRIN-6698 namespace (50/17)	0.835 ± 0.026	0.857 ± 0.029	+0.022 [-0.014, +0.063]

Table B.16: Archived seed44 epoch 20 fixed-training-probe diagnostics, shown as mean ± descriptive sample SD across five fold-specific statistics. Skill is $1 - \mathrm{L1}_{forecast}/\mathrm{L1}_{copy}$ within each snapshot's EMA coordinates; positive is better than holding the landmark state constant. Forecast targets require a continuous actually observed visit chain. These are training-probe summaries, not best-checkpoint holdout forecast evaluations. NoJEPA has no trained-world diagnostic.

Model	Horizon 1 skill	Horizon 2 skill	Horizon 3 skill
NoJEPA	–	–	–
DirectReadout	+0.026 ± 0.056	+0.050 ± 0.142	+0.193 ± 0.134
ForecastAttention	-0.037 ± 0.016	-0.015 ± 0.071	+0.024 ± 0.028

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